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MODAL ANALYSIS AND RESONANCE AVOIDANCE OF BLADED DISKS

Frank Kushner
Consulting Engineer
Frank Kushner Consulting, LLC
Delmont, PA, USA

Brian Pettinato
Manager, Aero and Structural Dynamics
Ebara Elliott Energy
Jeannette, PA, USA

Mark Pechulis
Mechanical Engineer, Solid Mechanics
Ebara Elliott Energy
Jeannette, PA, USA

Manish Thorat
Senior Engineer, Machinery Dynamics & Acoustics
Ebara Elliott Energy
Jeannette, PA, USA

Frank Kushner is a consulting engineer and expert witness specializing in dynamics and acoustics testing and analysis. He has had 54 years of experience with industrial turbomachinery, mostly all while with the Elliott Group, and four years previous experience with the jet engine combustion section development group at Pratt & Whitney Aircraft. After obtaining a B.S. degree (Mechanical Engineering, 1965) from Indiana Institute of Technology, Mr. Kushner received his M.S. degree (Mechanical Engineering, 1968) from Rensselaer Polytechnic Institute. He is a registered Professional Engineer in the State of Pennsylvania and a member of the ASME and the Vibration Institute.

Brian Pettinato is a Senior Manager of Aero and Structural Dynamics at Ebara Elliott Energy in Jeannette, Pennsylvania. He has been with Elliott since 1995. His primary area of expertise is machinery dynamics. Prior to joining Elliott, he worked as a project engineer for CentriMarc. Mr. Pettinato received his B.S. (1989) and M.S. (1992) Mechanical Engineering degrees from the University of Virginia. He has coauthored over thirty technical papers and holds four U.S. patents. Mr. Pettinato is a fellow member of ASME, a member of STLE, and a registered Professional Engineer in the State of Pennsylvania. He serves on the Turbomachinery Advisory Committee of Texas A&M and on the API 684 rotordynamics task force.

Mark Pechulis is a Development Engineer in the Aero and Structural Dynamics Department, Solid Mechanics Group at Ebara Elliott Energy in Jeannette, Pennsylvania. He has been with Elliott since 2013. His specialties include modal testing and acoustic measurements. Prior to joining Elliott, he worked as a Technical Leader in the Electro-Mechanical Testing group of GE Power and Water. Mr. Pechulis received his B.S. (1992) Mechanical Engineering degree from Western New England University and his M.S. (1994) Mechanical Engineering degree from Lehigh University.

Manish Thorat is a Senior Engineer and Machinery Dynamics & Acoustics Group Lead, R&D at Ebara Elliott Energy. He has been with Elliott since 2010. His areas of expertise include rotordynamics, vibration analysis, testing and evaluation of seal designs, squeeze film dampers and fluid film journal bearing design. Mr. Thorat holds two U.S. patents and has co-authored five publications in ASME journals. Mr. Thorat is a member of ASME and STLE. Mr. Thorat earned his MS degree in Mechanical Engineering from Texas A&M University (2010) and BE in Mechanical Engineering from Mumbai University (2007).

ABSTRACT

Rotating disk and blade fatigue failures represent a low percentage of failures in industrial turbomachinery, but other than coupling / shaft end failures remain some of the most problematic for extensive repairs and lost availability. This tutorial examines the different sources of bladed disk excitation and their potential to induce resonant excitation. Fluid-structure-interaction computational methods

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and modal testing are further discussed and used to improve overall understanding including statistical avoidance of resonant excitation. Information on the effects of mistuning of blades and disks, some beneficial and some with serious concerns for increased resonant amplification is further presented. Finally, additional knowledge regarding acoustic pressure pulsation excitation, especially for centrifugal impellers at rotating blade passing frequency is covered through a review of acoustics calculations for failure prevention.

INTRODUCTION

High cycle fatigue (HCF) of bladed disks remains one of the most critical issues for fluid-handling turbomachinery for the oil, gas, petrochemical, and related industries. As reported by the International Association of Engineering Insurers (2005), a study by the North American Electric Reliability Council (NERC) and EPRI indicated that LP turbine blades were among the largest contributors to Steam Turbine non-availability (Figure 1). These failures were primarily due to stress corrosion cracking and excessive erosion; however it was noted that "There continue to be resonance issues/failures with steam turbine blading. While many of the problems with older turbine designs have been resolved or managed, some of the new turbine designs for either small or large units have had cracks/failures particularly with the last stage blades."

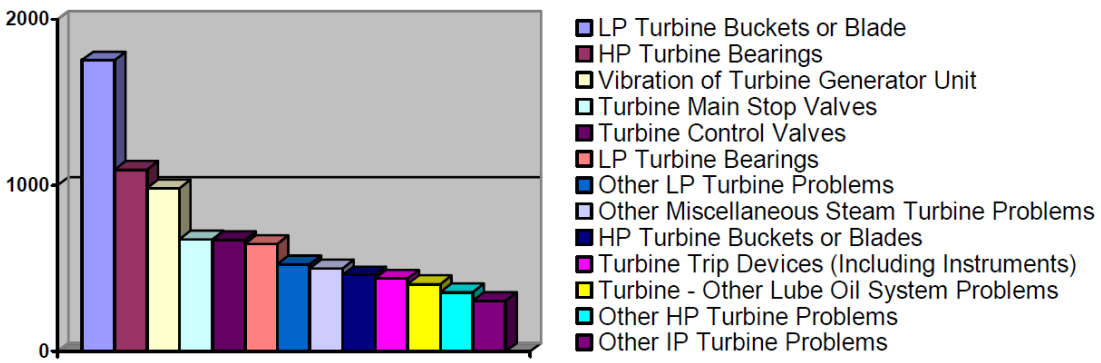

Figure 1. Ranking of Top 15 Failure Causes for Fossil Steam Turbine Lost Availability in MW-Hrs per Year from 1998-2002. Courtesy NERC and EPRI, (IMIA, 2005)

There are also similar issues with jet engine designs, their land-based derivatives, and other gas turbines including microturbines (Srinivasan, 1997). While these applications have much different operational modes and life cycles (Layton and Marra, 2000), reports attribute between 40 and 50 percent of gas turbine outages to blade fatigue. Gas turbines have some causes of blade fatigue that are unique to their architecture such as combustion pulsation, which gives rise to alternating pressure forces. Gas expanders such as those for fluid catalytic cracking (FCC) processes are a special case with the combustion occurring in the process, but there have been few blade fatigue problems in expanders that were not associated with some other issue such as corrosion, described by Dowson, et al. (1995), or catalyst dumps into the expander inlet casing.

A similar survey was provided for a broad range of machinery focusing on petroleum refining, chemical, petrochemical, and natural gas transmission industries (Clark, 1996). One insurer's experience was that centrifugal, axial, and rotary compressor losses represented the largest portion of rotating equipment losses. Among those losses, the fatigue and vibration category was the third leading cause. There is, of course, a severe monetary impact to equipment failure in the power and process related industries.

This tutorial will focus on the avoidance of fatigue related failures particularly as it pertains to resonance avoidance and operation within resonance. Some of the materials to be presented include

- Historical development
- Design methodology
    - Disk critical speed avoidance
    - Cantilevered blade design and resonance avoidance
    - Impeller design and resonance avoidance
    - Operation in resonance
    - High cycle fatigue
- Case studies

Root cause of failure cases will be described. These will include two cases for axial compressors, and several cases for single and multistage centrifugal compressors along with several cases for steam turbines. This tutorial will present some important views on how components should be designed so that they meet the service requirements for the required life (20 years per some specifications with 5-year runs between checks). Of course components also must be fabricated with specified materials, inspected to show they conform

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to design concepts, and operated and maintained properly. Whenever there are failures even from mis-operation, it is always wise to use the knowledge gained to achieve improved designs for "fitness-for-service."

HISTORICAL DEVELOPMENT WITH SOME EXAMPLES

Research into bladed disk vibration and failures has been going on for over one hundred years. One of the early works is Stodola's 5th edition of Steam Turbines, a German edition that does not appear to have been translated into English. Wilfred Campbell was apparently well aware of this work having referred to it in his paper.

Nodal Diameters

The vibration of membranes and disks were first studied in relation to sound. Ernst Chladni (1802) performed sand experiments to determine the nodal lines of both simple and complex structures including disks. This would mostly involve drawing a violin bow at the edge of the structure. As the structure would vibrate, sand (or flour) would bounce off the structure except at nodal line configurations where the sand was not moving. Figure 2 shows some nodal patterns that were determined by experiment by Chladni. Figure 3 show similar nodal patterns that were determined experimentally for turbine disks. In the case of the turbine disks, an electromagnetic shaker was used to provide the vibration. The point where the excitation was applied, either by the bow or by the shaker was always located at the mid-point between nodal lines.

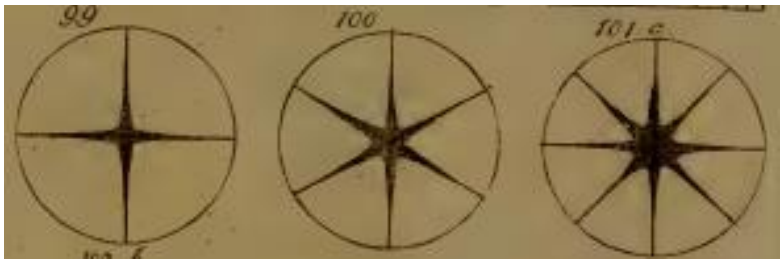

Figure 2. Disk nodal patterns (Chladni, 1802)

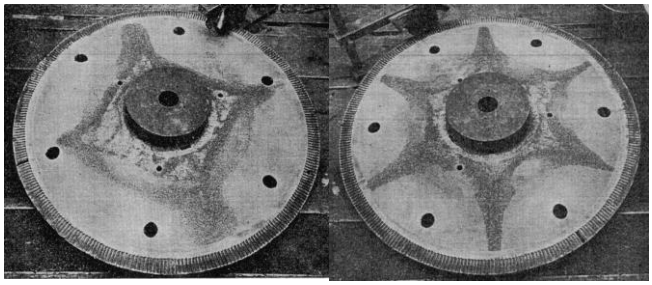

Figure 3. Turbine disk nodal patterns (Campbell, 1924)

The theoretical equations governing the non-rotating circular disk with a free boundary condition were eventually solved by Kirchoff in 1850. Armstrong, et al. (1965) provides Equation (1), which relates the static natural frequency of a disk to its variables. The natural frequency varies in proportion to the thickness of the disk and inversely to the square of the radius.

$$f_s = \frac{\eta}{4\pi} \cdot \frac{t}{r^2} \cdot \sqrt{\frac{g_c \cdot E}{3 \cdot (1-\nu^2)}} \qquad (1)$$

such that for a particular mode,

- f_s = *natural frequency of the disk under static conditions (Hz)*
- η = *constant that is a function of the number of nodal circles (s), nodal diameters (n), and boundary conditions.*
- t = *disk thickness*
- r = *disk radius*
- g_c = *gravitational constant*
- E = *modulus of elasticity*
- ν = *Poisson's ratio*

The problem of a turbomachinery disk is of greater complexity since (i) there is a bore that is fixed and (ii) the cross-section is not uniform throughout. The natural frequencies and modes of such complex structure can be determined by either testing or by finite element analysis (FEA), which will be discussed in greater detail later in the paper.

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Centrifugal Stiffening

Lamb and Southwell (1921) discussed forward and backward traveling waves in a spinning disk and also determined the relation for natural frequencies in rotation, $f_r^2 = f_s^2 + f_c^2$, which is expressed in its more commonly used engineering form as

$$f_r = \sqrt{f_s^2 + \beta \cdot N_s^2} \tag{2}$$

such that for a particular mode,

f_r = *natural frequency of the spinning disk*
f_s = *natural frequency of the disk under static conditions*
β = *stiffening coefficient*
N_s = *speed of rotation*

Disk Critical Speeds

Wilfred Campbell (1924) provided a treatise on the axial vibration of steam-turbine disk wheels. Of particular importance was the actual measurement of forward and backward traveling waves around the circumference of the disk using stationary and rotating coils. Many disk failures were known to have occurred when the speed of the backward traveling wave was equal to the rotational speed thereby creating a stationary wave. Some disks vibrated so violently in the axial direction that the bladed disks reached adjacent stators and rubbed. This situation has been described as a disk critical speed or major wheel resonance and is provided in Equation (3).

$$N_{cr} = \frac{f_r}{n} \tag{3}$$

such that

N_{cr} = *disk critical speed corresponding to the n-diameter mode*
n = *number of nodal diameters*
f_r = *natural frequency of the wheel pertaining to n-nodal diameters at rotational speed* $N_s = N_{cr}$

One item that is conspicuously absent from the disk critical speed equation is a source of excitation. The source of excitation was determined to be due to "windage effect". In essence, the gas itself was causing and sustaining the vibration, and the vibration would disappear as vacuum was pulled. In turbomachinery, small circumferential variation in the pressure is sufficient to cause excitation of disk critical speeds. This variation can come from non-uniform flow entry as encountered at inlet stages, nozzle spacing variation, and other small geometric variations in construction. In general, the first four disk critical speeds were to be avoided.

During the time of Campbell's publication, designs were being pushed to their mechanical limits with respect to stress. This led to the thinning of disks for bore stress reduction and also for weight reduction to better control rotor critical speeds. This trend of disk thinning eventually led to an increased number of disk failures due to disk critical speeds. These failures are generally not encountered today as all reputable turbomachinery manufacturers avoid disk critical speeds. Figure 4 shows a disk fatigue failure on the 11th stage of a 30 MW 1500 rpm steam turbine. One can take special notice of how thin the disk is.

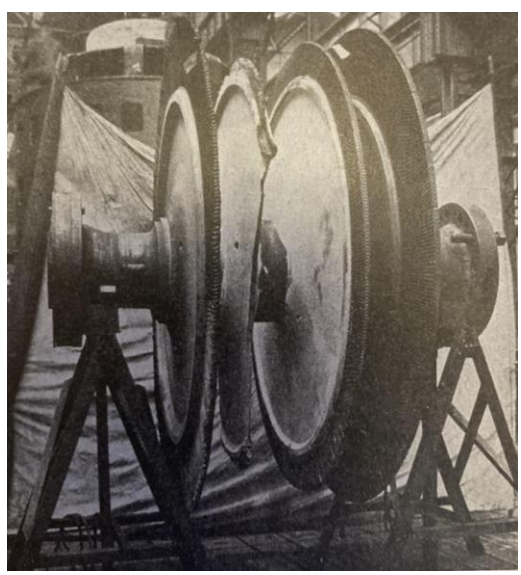

Figure 4. Fatigue failure of a turbine disk (Campbell, 1924)

Bladed Disk Natural Frequencies

The free standing blade natural frequency and bending mode are treated as being analogous to that of a straight bar or "reed" where the blade is free to vibrate above the base of the blade. Turbine blade cross-sections create a weak axis (typically coinciding with tangential

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motion) and a strong axis (typically coinciding with the axial direction), and the natural frequency must be determined in both directions.

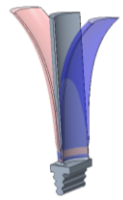

Figure 5. Blade 1st mode

The first natural frequency of a cantilever beam (Figure 5) at non-rotating stationary conditions is described in Equation (4)

$$\omega_n = 3.52 \cdot \sqrt{\frac{(E \cdot I)/l^3}{\mu_1 \cdot l}} \tag{4}$$

such that

- ω_n = *the blade natural frequency (rad/s)*
- E = *modulus of elasticity*
- I = *area moment of inertia*
- μ_1 = *mass per unit length*
- l = *length of the blade*

Natural frequencies of short blades was noted to be particularly problematic since the blades are so stiff that the disk rim has a distinct effect such that the natural frequency is essentially that of the blades and disk rim. This natural frequency asymptotically approaches the blade only natural frequency as the number of nodal diameters increases.

Campbell published his second paper on bladed disks in 1925. This time he tackled the subject of tangential vibration of steam-turbine buckets (Campbell and Heckman, 1925). The "frequency-speed diagram" was utilized to diagnose potential problems (Figure 6). This diagram would later be dubbed the "Campbell diagram". Campbell noted that the common practice in Europe at the time was to avoid the use of buckets having a natural frequency of less than 3.5 times running speed. Based on test results, he concludes by advocating avoidance of the 2nd, 3rd, and 4th harmonics of running speed noting that ". . . the amplitudes of motions for the 5, 6, and 7 points are so small that we ignore their location." It is of some interest that avoidance of the nozzle passing frequency is not mentioned; however, failure prevention of long blades was the focus of the paper and the natural frequencies of these blades were being encountered well below the nozzle passing frequency.

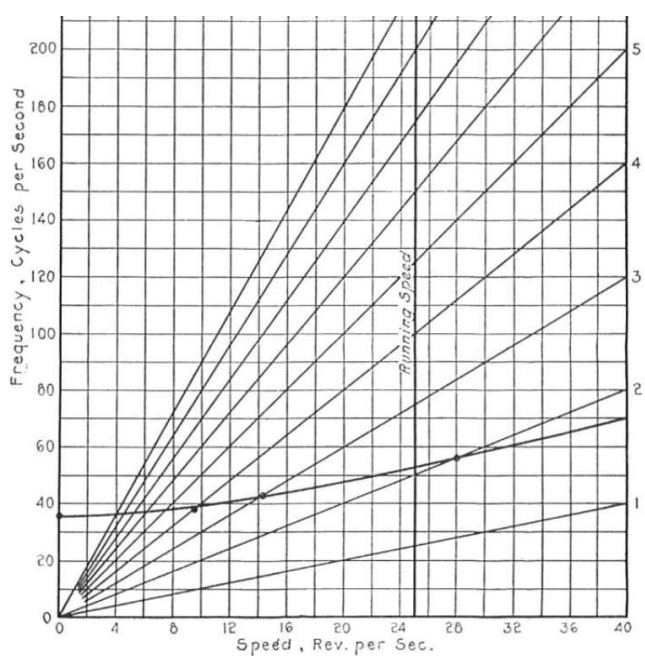

Figure 6. Frequency vs speed diagram showing the first tangential mode (Campbell, 1925)

Avoidance of the tangential natural frequencies can be attained by deliberate design and/or by tuning. The essential equation for any natural frequency is defined by Equation (5)

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$$\omega_n = \sqrt{\frac{K}{M}} \tag{5}$$

such that

ω_n = *natural frequency of the mode of interest*
$\boldsymbol{K}$ = *modal stiffness*
$\boldsymbol{M}$ = *modal mass*

Adjustment of the natural frequency becomes a matter of manipulating the stiffness and mass of the blades. In Campbell's time, increasing the natural frequency was performed by stiffness increase using shroud bands and/or lacing wire as stiffening devices that would group blades together. The importance of lacing wire as a source of frictional damping was not yet realized at the time. Natural frequency could also be increased by material removal from locations that did not affect stiffness. Decreasing the natural frequency was performed by stiffness reduction from removing material from the backs of buckets.

The engineering community has taken these early methods and ran with them toward logical and some illogical conclusions. One practice for raising the natural frequency that has been abandoned is the circumferential joining of blades together at their tips by weldment and the use of welded intermediate platform locations for the sake of stiffening. The centrifugal and thermal forces are such that the blades want to grow outward in diameter and circumference. Excessively restricting that movement has led to fractures in shrouds and blades.

Partial Arc Admission

Kroon (1940) reported on the effect of partial admission, which at the time of his publication, was producing an industry-wide problem of shroud and blade failures. This was recognized to be a timing problem in that a large stimulus is applied when entering and leaving the arc of admission. If precisely timed, the forcing function can be in-phase with the blade or packet vibration with only damping to limit the amplitude. By assuming the worst possible timing of the load, the oscillating stress can be based upon the static steam stress times a function of the logarithmic decrement and cycles of vibration per revolution as follows:

$$\sigma_v = \sigma_s \cdot \left[1 + \frac{2}{n \cdot \delta}\right] \tag{6}$$

such that

σ_v = *vibratory bending stress*
σ_s = *static steam bending stress*
n = *cycles of vibration per revolution*
δ = *logarithmic decrement*

The length of the arc of admission is negligible and the stresses are in essence inversely proportional to the damping with the obvious implication that no damping will lead to infinite stress. Measurement of material damping was just being started at this time and results for a few materials were presented. More sophisticated examinations of partial arc admission would be provided later by Pigot (1980) and Rieger (1990).

Resonant Response

Prohl (1958) and Weaver and Prohl (1958) provided an analysis method for banded groups of turbine buckets, a problem that was described by one of the reviewers of his paper as having ". . . occupied the efforts of numerous investigators." The equation for blade stress becomes

$$\sigma_v = \sigma_s \cdot S \cdot R \cdot \left(\frac{\pi}{\delta}\right) \tag{7}$$

such that

σ_v = *vibratory bending stress*
σ_s = *static steam bending stress*
S = *stimulus multiplier*
R = *resonant response factor*
$\frac{\pi}{\delta}$ = *amplification factor*

This simple equation sets the stage for acceptable operation in resonance. The key items to being able to do so are the accurate determination of each of the variables in Equation (7) and an accurate determination of the vibratory stress limit. Those key items are of such significance and complexity that they have been the focus of considerable research since Prohl's publication. Prohl devoted much of his paper to the derivation of resonant response factors for banded groups of blades. The resonant response factor, R, is affected by both the mode shape and response to the force based on phase angle. Thus for an unshrouded cantilevered blade, R would be close to 1.0, while for an optimized packet of blades, R can usually be reduced to roughly 0.2 or less for the fundamental modes. The

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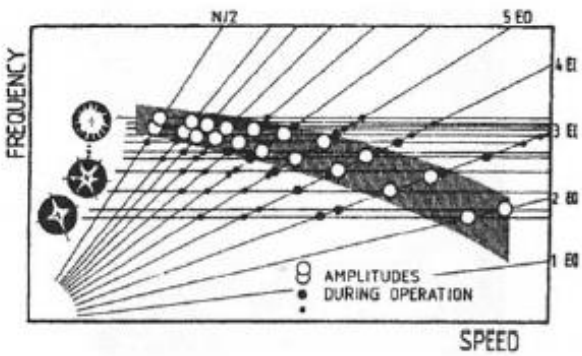
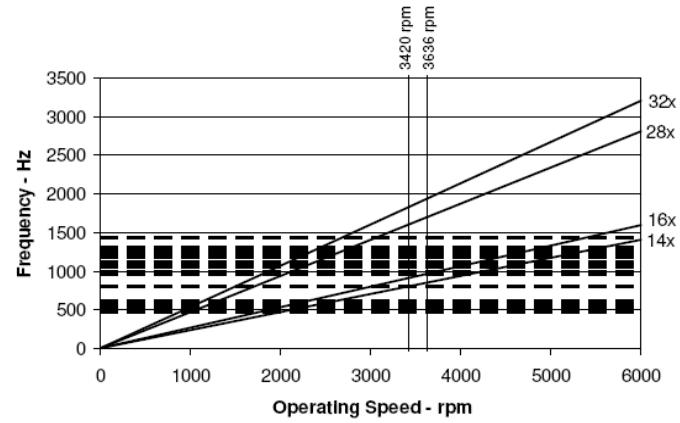
calculation of resonant factor for blade packets having in-phase mode is provided in Appendix A.

Rotating Blade / Stationary Vane Interaction Resonance

Applying the Campbell diagram to an entire disk tends to yield the situation shown in Figure 7 showing many modes having interaction resonance. There always seems to be some mode coinciding with some multiple of operating speed. It is simply not possible to avoid everything. When considering the successful operation of many bladed wheels without failure, it becomes self-evident that it is not necessary for every mode to avoid every multiple of running speed and vane pass. This invites the questions: Which modes are important to avoid? Which modes are not? And what excitation frequencies matter?



(Pfeiffer, 1985)



(Pettinato, et al., 2013)

Figure 7. Campbell diagram of bladed disks having unavoidable minor interaction resonance

Prior to the 1970's it was perplexing when "minor resonance" could lead to disk fatigue failures. There were occasional unexplained albeit rare failures of compressor impellers. In 1974-1975, strain gauges on impellers were used to solve recurring failures at an LNG plant related to a 27 bladed impeller downstream of 16 return channel vanes. What is termed interaction resonance of bladed disks was discovered. The renowned vibration expert, MIT's Prof. Den Hartog, since deceased, had met with the parties and other consultants before strain gauges were finally applied with slip ring data transfer. He noted that it was the most difficult problem he ever experienced. There were multiple impeller failures. Two different crack patterns were exhibited. After months of testing and removing the cause of high excitation, which was severe liquid ingestion, the question remained: what about future designs? While reviewing data the puzzle was solved and response equations were developed to prove what "minor resonances" could really be important. The equations verified why the 5-diameter disk mode responded at a frequency equal to two times number of inlet vanes, while the 3-diameter mode did not respond to even higher excitation forces at one times the number of vanes. Strain gauges were essential to the discovery and also explained the other crack pattern was due to loss of aerodynamic damping causing flutter of the 2-diameter mode when in compressor overload with liquids.

Kushner (1979, 1980) was allowed to publish his work and provided a set of parametric equations for evaluating interaction resonance (also known as minor resonance) using the number of rotating blades and stationary vanes to determine the excitation and response potential. For certain combinations of the two numbers, the coupled blade/disk modes with diametric nodal lines are either phase cancelled or excitable when an excitation source coincides with a resonant mode. The equations required for checking for disk mode "interaction resonance" and "disk critical speeds" are as follows:

When not at a disk critical speed:

$$n = |y \cdot s \pm z \cdot b| \quad (8a)$$
$$y \cdot s = h \quad (8b)$$
$$f_r = y \cdot s \cdot \omega \quad (8c)$$

When at a major disk critical speed:

$$\text{For } b > 1 \quad (9a)$$
$$y \cdot s = h = n \quad (9b)$$
$$f_r = n \cdot \omega \quad (9c)$$

where

b	= number of rotating blades	h	= harmonic of speed or engine order (EO)
s	= number of stationary elements	n	= number of diameter nodal lines
f_r	= natural frequency at speed, Hz	y & z	= integers > 0
		ω	= rotating speed, Hz

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Modes that meet either Equation (8) or Equation (9) are excitable. By contrast, modes that fail to meet these equations experience considerable phase cancelation and are relatively difficult to excite. Modes of concern range from the 0-diameter mode up through the number of blades divided by two-diameter modes. The number of stationary vanes should never be equal to the number of rotating blades to avoid exciting the 0-diameter (ND0) mode. This combination is known to cause high noise regardless of the separation margin, and the torsional mode of the cover to disk also becomes a concern. The one-diameter disk mode (ND1) is usually coupled to the shaft and is not easily excited and therefore typically not a concern even when Equations (8) and (9) are satisfied. To date, the plus sign in Equation (8a) is not important for radial compressors; it results in high values of "n" diameters, difficult to excite, let alone find in testing. Equations (8) and (9) are often represented graphically by the interference diagram, which is also called a ZZENF diagram (Wildheim, 1979) or "SAFE diagram" (Singh, 1988). The aircraft industry terms this phase cancellation as "aliasing". Such methods are used to determine whether or not a certain number of stator vanes could excite a well-tuned impeller or blisk to resonant vibration in its primary modes.

An interference diagram is constructed in Figure 8. The impeller has 13 blades and is fed by 14 effective sideload guide vanes (SLGV) and 16 return channel vanes (RCV). Lines of excitation are drawn starting from the origin and extending to the untruncated B/2 nodal diameter, which for this 13 bladed impeller is 6½. The slope is equal to the harmonic frequency such that the 1st intersection with ND1 is 1x running speed and the 1st intersection with ND2 is 2x running speed, etc. The first sloping line defines the disk critical speeds. Once the first segment reaches the b/2 nodal diameter, it starts a second segment having a negative slope that progresses upward when moving to the left. The 2nd intersection with the ND6 is at 7x running speed and the 2nd intersection with ND0 ix b x running speed or 13x running speed. This pattern is repeated in a zig-zag fashion extending up to or past the highest 2S harmonic, which is 2x16RCVs or 32. Terminating this final segment at the end terminates at 6½-ND and 32½ harmonics. All of the nodal diameter modes are plotted on the diagram. The modes of concern are shown in red corresponding to the 1xSLGV, 1xRCV, 2xSLGV, and 2xRCV. The red bars coincide with maximum continuous speed (N_{mc}), minimum continuous speed (N_{ma}), 2xN_{mc} and 2xN_{m}a, 10% above and below these values. All other modes are not of concern as they would experience phase cancellation.

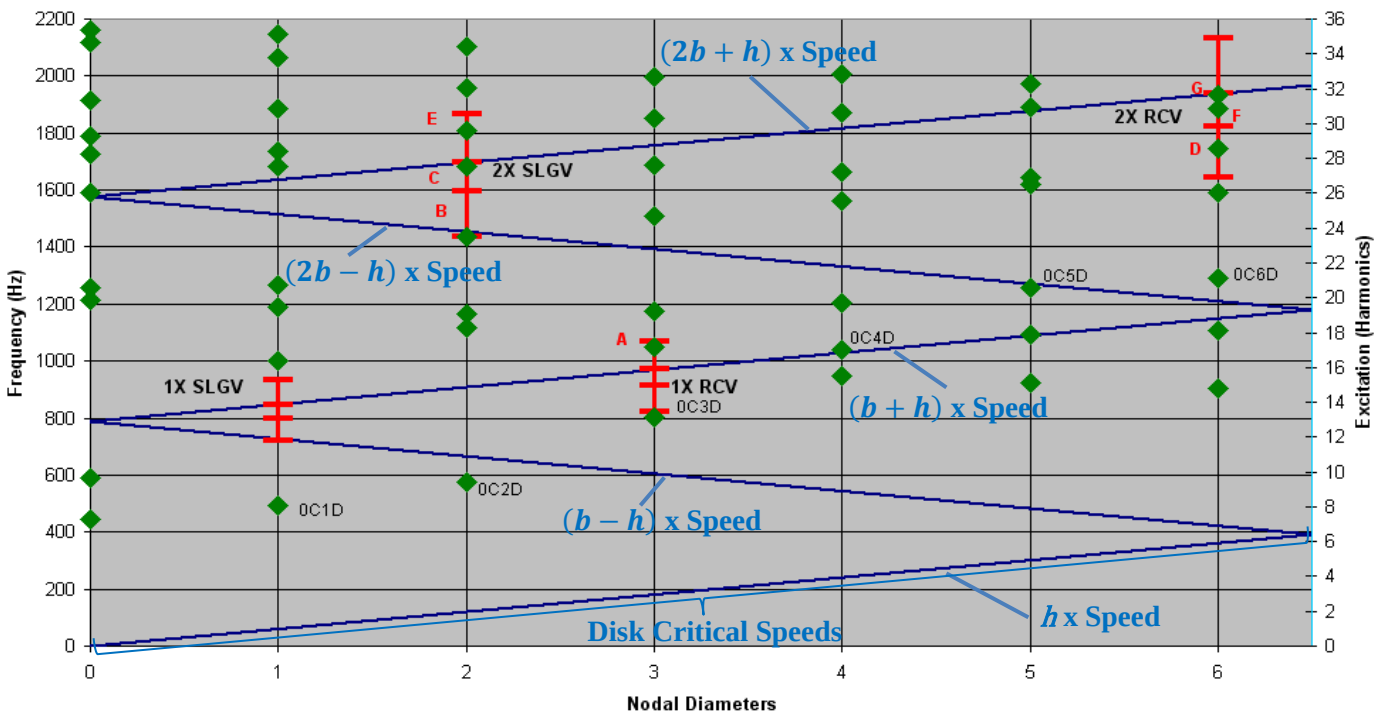

Figure 8. Interference diagram for a 13 bladed impeller

The effectiveness of this aliasing technique was demonstrated by Pettinato, et al., (2013). The response of the ND3 disk mode was determined for different excitation patterns sweeping through the ND3 natural frequency. Figure 9 shows the results where 13x through 19x patterns were applied. In essence, only the 16x pattern was shown to provide high excitation of the 13 bladed impeller's ND3 disk mode. Response of the other excitation patterns were small by comparison and roughly analogous to having good separation margin.

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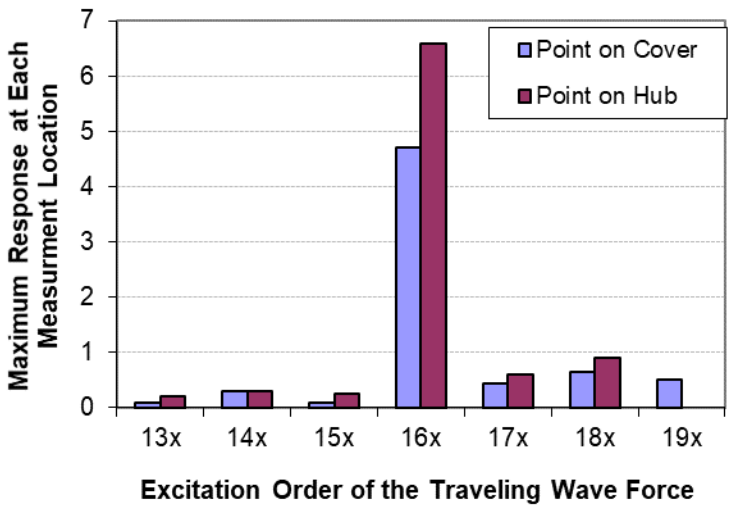
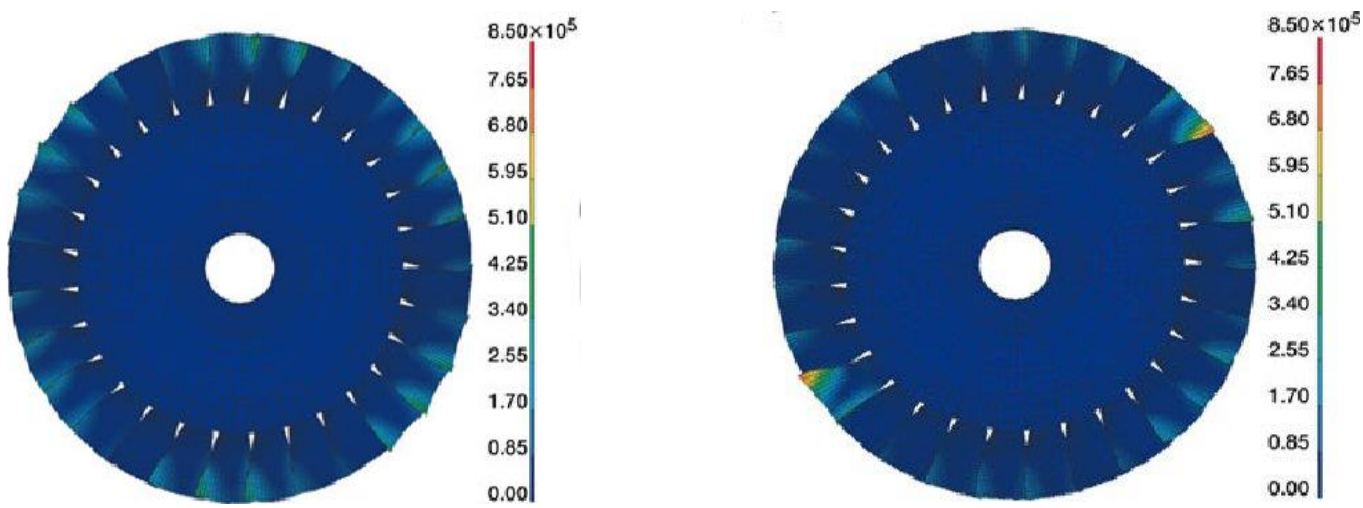


Figure 9. Maximum response of the ND3 disk mode of a 13-bladed impeller.

Mistuning

Mistuning refers to the fact that turbomachinary components, such as fan, compressor and turbine blades, are not identical, but have slightly different frequencies. The frequency differences can be the result of material or dimensional variation inherited from manufacturing tolerance and assembly differences. Corrosion, erosion, deposits, or foreign object damage during machine operation can also increase these frequency differences (Kushner, 2004, McCloskey, 2002). While the advances in design and manufacturing techniques have been able to produce better bladed systems with increased uniformity, i.e. the frequency differences are very small, smaller mistuning does not necessarily translate into smaller amplification factors in the forced vibration cases.

The problem with mistuning lay in the fact that it can potentially lead to large local vibration level around one or a limited number of blades, increase the stresses of blades compared to the perfectly tuned system, thereby increasing the likelihood of a high cycle fatigue failure. Figure 10 shows the displacement for cantilevered blades on a disk for both the tuned and mistuned cases. With the blades all tuned there is a 3-diameter pattern. But some blades can have much higher response when there is a mistuning of blades as shown in Figure 10 for the same resonant frequency.



(A) Tuned (B) Mistuned

Figure 10. Tuned and mistuned three-diameter disk mode

The mistuning level is provided by Equation (10)

$$\frac{f_i - f_{avg}}{f_{avg}} \tag{10}$$

where f_i is the modal frequency at different locations or blades and f_{avg} is the average modal frequency.

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For forced excitation, Bladh, et al. (2002), show on a specific disk that mistuned responses can exceed tuned response levels by nearly 200 percent, if appropriate levels of mistuning and interblade coupling are present. Fortunately, this is a statistical anomaly representing the 99.9th percentile worst response at the worst possible condition. The 95th percentile would yield 1.8x the tuned. In the process equipment industry, mistuning is rarely taken into consideration, generally being covered by "proven" safety factors. Whitehead (1966, 1998) found that the maximum increase in vibratory response that you can get from mistuning is proportional to $\sqrt{N}$, where N is the number of blades on the rotor. Since impellers generally have fewer blades than axial flow turbomachinery, they have lower potential for vibration increase due to mistuning. Pettinato, et al. (2013) performed mistuning measurements on a 13-bladed impeller. The amplification magnification due to mistuning could be as much as 1.48, a value that is typically covered by most safety factors.

Mistuning also affects the ability for phase cancellation to take place. Within the mistuning community, phase cancellation is known to work when mode pairs split, but are not distorted. When the mistuning is high enough, the mode of concern can interact with other nodal diameters, which can impair the desired phase cancellation. The mistuning level of a 13 bladed impeller is provided in Figure 11. This ND3 mistuning level was only 0.01%. Blade mistuning levels were higher at 0.5%. Since the next nearest mode for this case was 8% away, phase cancellation of the type described by Kushner (1979) can be relied upon.

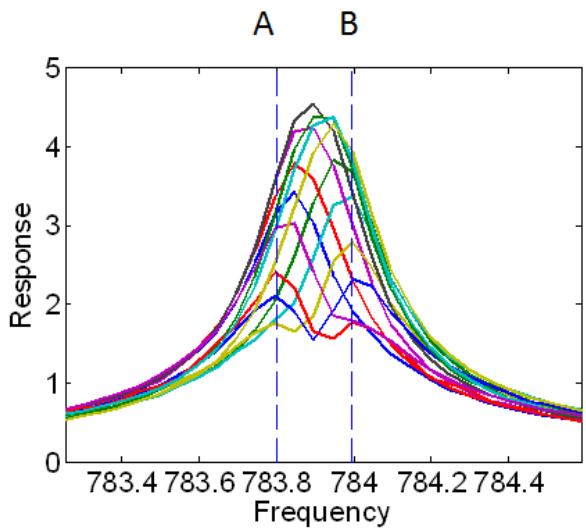

Figure 11. Zooming to show response of 3-diameter mode measurements taken at different locations

Although mistuning can be harmful to a machine, it is not always so. Whitehead (1966) and Ewins (1969) published the first studies of mistuning effect on flutter stability and forced response. In many cases, it was found that mistuning can have a beneficial effect on blade flutter stability in gas turbines. The use of intentional mistuning of bladed disks to reduce their sensitivity to unintentional random mistuning has been well adapted in aircraft gas turbine engines.

DESIGN PHILOSOPHY

The basic design practices that were developed decades ago are still applied today. At this point, the design prevention of bladed disk failures consists of (i) avoiding disk critical speeds, (ii) avoiding excitable natural frequencies in the operating speed range, which is typically limited to fundamental and low order modes, and (iii) keeping stresses below an acceptable vibratory stress limit. In actuality, it is all about keeping the stresses below a stress limit, it's just that avoiding resonance is often considered sufficient in and of itself to achieving this goal. A bladed disk with cyclic stress below its modified endurance limit will theoretically provide infinite life within the continuous operating speed range whereas exceeding the vibratory stress limit will reduce the safety margins and create a risk of high cycle fatigue.

Disk Critical Speed Avoidance

The most destructive resonance is that for disk critical speeds. Disk critical speeds are rarely encountered as most disks are simply thick enough and rigid enough to push these modes above the operating speed range. In addition, all turbomachinery manufacturers avoid them. Very thin disks such as may be found in uncovered semi-open impellers can be particularly vulnerable especially if manufactured from titanium. Figure 12 provides a useful illustration of the fundamental two-diameter mode shape of a covered impeller. Sections of the hub disk move along with the cover disk. This is a mode for the axial direction; the 15 blades also have relative tangential motion. There are two opposite areas in-phase; simultaneously two areas are out-of-phase. There are four radial nodal lines (two nodal diameters) with zero motion.

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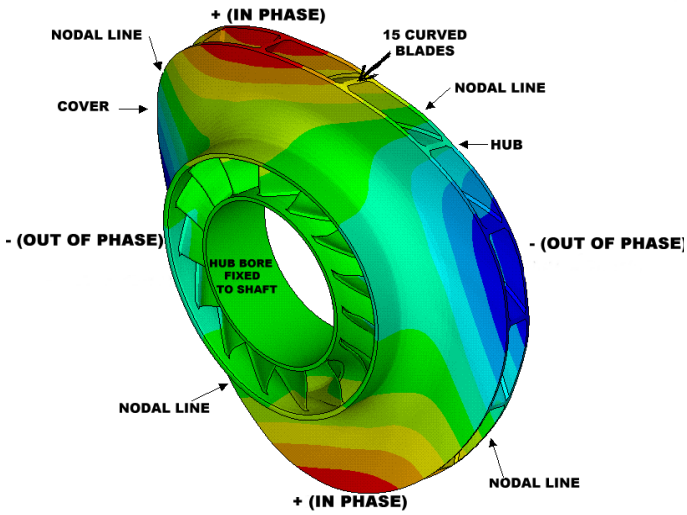

Figure 12. Mode shape of two-diameter disk mode for covered impeller

Assume the 2-diameter mode frequency of a disk would be 200 Hz when at 6000 rpm. As this disk rotates, at a certain speed each part of the disk would arrive at an area in-phase with a stationary exciting force. As shown in the Campbell diagram (Figure 13) at 6000 rpm, resonant operating speed is equal to the natural frequency divided by the number of diametrical nodal lines.

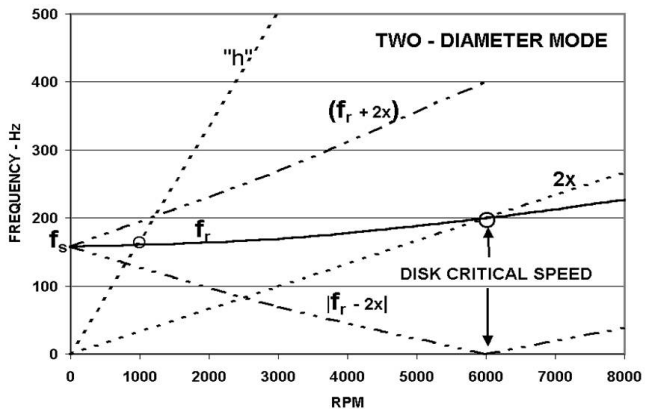

Figure 13. Campbell Diagram for Disk Critical Speed.

In Figure 13, the "backward wave" line, $|f_r - 2x|$, intersects the x-axis when the disk frequency is equal to two per revolution. Note that f_r, frequency at speed, includes effect of temperature, fluid mass loading, and centrifugal loading on stationary frequency, f_s.

Cantilevered Blade Design and Resonance Avoidance/Mitigation

Free standing blades are the most at risk for HCF as they are the most compliant, not only in axial compressors but also in non-shrouded stages of both steam turbines and centrifugal compressors. Free standing, cantilevered blades come in many forms and sizes. Shorter blades usually do not have much twisting along the profile length and have three fairly independent modes: fundamental weak-axis (tangential), fundamental strong axis (axial) and fundamental torsional. Higher modes have nodal lines and have phase changes along the length and width and thus have inherent phase cancellation for most cases. The fundamental weak axis mode is not only the most compliant, but typically experiences the highest excitation forces as the direction of force (a percentage of steady forces) aligns with the predominant steady gas force that "drives" the machine. Many blades are canted so that gas forces excite both the weak and strong axis of bending. However, excitation is greatest about the weak axis and smallest about the strong axis. In addition, the strong axis is less compliant giving lower dynamic stresses. Torsional modes also are of lower concern, as there is a node line along the blade, so that the leading edge exciting force is nearly out-of-phase with that on the trailing side. For short blades, axial and torsional modes can be at least less than one tenth of the concern for tangential bending modes. There could of course be exceptions, such as from axial forces on a highly loaded reaction-bladed turbine stage.

Medium length blades such as the middle stages of steam turbines typically have much fewer problems as discussed in the introduction. Nozzle passing frequency is avoided; resonance of the first modes would only be equal to lower harmonics of speed where severe non-uniformity of nozzle throats is the main issue; along with potential excessive corrosion in transition stages (Case C-1 presented later). Higher order modes could also present problems with excessive corrosion.

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Longer cantilevered blades such as those in Figure 14 in the rear of steam turbines and also those without friction damping in the front-end of axial compressors can have resonance potential for all three directions. In modern machines, longer blades have significant twist and have larger chord widths; thus the modes for the three directions are coupled. In fact there can be higher axial motion as compared to tangential at the blade tip for the lowest frequency mode; thus higher axial forces for a reaction-type stage of a steam turbine could give relatively more response at resonance. In addition, the fundamental modes also have potential for flow-incidence related flutter. The longer freestanding blades in many steam turbine-generator exhaust ends usually have the first few modes carefully tuned. Gas expanders should have less risk, especially since inlet connections are axial reducing harmonic excitation, other than from support struts.

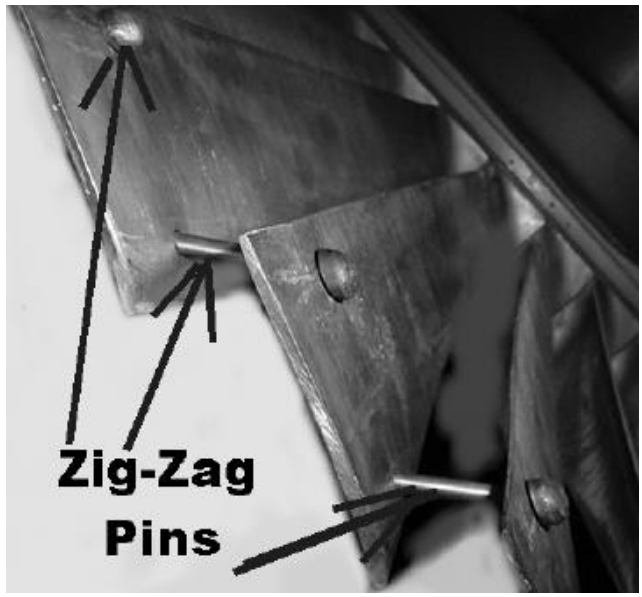


Figure 14. Turbine last stage blades with zig-zag damping pins

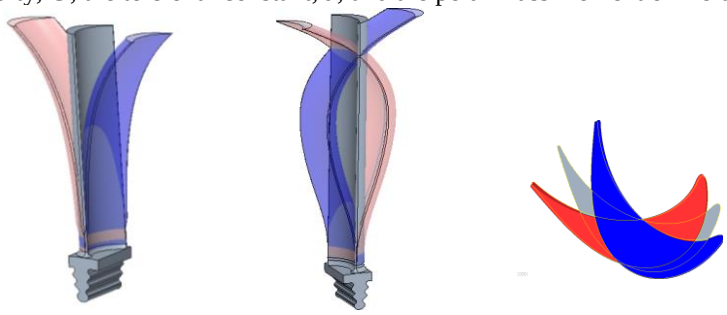
Blading designed to API 612 8th edition must meet one of the following requirements:

- Resonance shall not occur within 10% of relevant excitation frequencies
- If any resonance is possible, the relevant dynamic stresses for any specified operating condition shall be low enough to ensure continuous, trouble-free operation as determined from dynamic stress calculation and comparison with allowable limits.

In this case, the modes to be considered include the tangential, axial, and torsional modes of individual blades as well as in-phase and out-of-phase packet modes all corrected for operating temperature and operating speed. The excitation frequencies to consider are

- First and second passing frequencies of upstream stationary blade rows
- First four turbine speed harmonics

Figure 15 provides mode shapes and equations for straight free standing blades. Variables for bending were described previously. Twist variables include the modulus of rigidity, *G*; the torsional constant, *J*; and the polar mass moment of inertia per unit length, *i*.



First Bending $\omega_n = 3.52 \cdot \sqrt{\frac{E \cdot I}{\mu_1 \cdot l^4}}$

Second Bending $\omega_n = 22.0 \cdot \sqrt{\frac{E \cdot I}{\mu_1 \cdot l^4}}$

Twist $\omega_n = 1.57 \cdot \sqrt{\frac{G \cdot J}{i \cdot l^2}}$

Figure 15. Blade bending modes

Shrouded Turbine Blades

Joining turbine blades together into packets by shrouding changes the natural frequency of each individual blade. In addition, the packeting of blades creates groups of motion as shown in Figure 16, which allows for phase cancellation of the packeted group. In this

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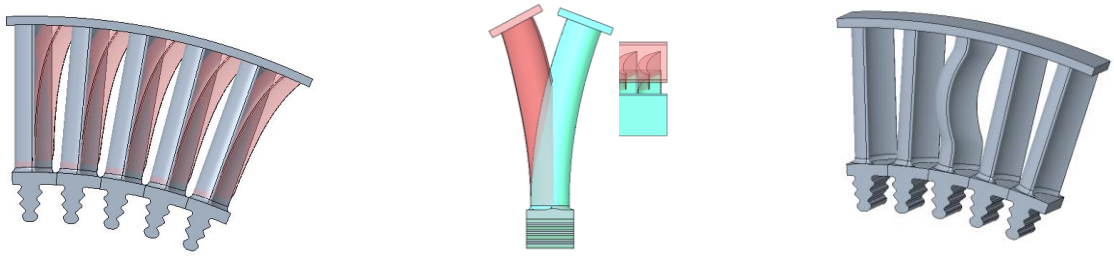
case, a_1 is a function of the shroud thickness. The design is such that a_1 is typically greater than 1.00 such that the first tangential natural frequency can be made higher than the free standing blade due to stiffening from the shroud.



Fundamental Tangential

$$\omega_n = a_1 \cdot \sqrt{\frac{3\cdot(E\cdot I)/l^3}{0.23\cdot\mu_1\cdot l + m_{shroud}}}$$

Packet Axial Translation

$$\omega_n = \sqrt{\frac{3\cdot(E\cdot I)/l^3}{0.23\cdot\mu_1\cdot l + m_{shroud}}}$$

Individual Bending in a Packet

$$\omega_n = 22.0 \cdot \sqrt{\frac{(E\cdot I)/l^3}{\mu_1\cdot l}}$$

Figure 16. Packeted modes

Phase cancellation has been used for many years in steam turbines and other machines to mitigate resonance. As discussed earlier, both Prohl (1958) and Weaver and Prohl (1958) give one of the best reviews on this subject. By tying turbine blades into packets, phase cancellation can be optimized for a given mode, so that some blades in a packet are out-of-phase with the exciting forces. Figure 17 and Figure 18 show how forces (F) can either line up or be out-of-phase with motion.

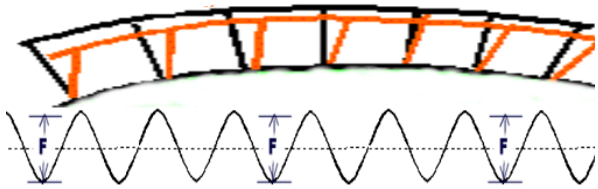


Figure 17. Blade packet fundamental mode all blades in-phase with excitation force (F)

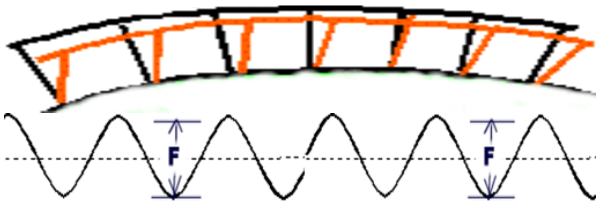


Figure 18. Blade packet fundamental mode: Some blades in-phase with excitation force (F); Some blades out-of-phase with excitation.

Phase cancellation is likely the most important reason over the years for reducing the number of HCF incidents in steam turbines. Response from partial-admission inlets causing non-uniform loading is also greatly reduced by shrouding. APPENDIX A gives a simplified method to check for optimum blade number per packet for the fundamental mode where all blades vibrate in-phase with the same amplitude. Similar procedures, considering amplitude and phase changes, can be done for higher modes, as shown for Figure 19. An example is shown in APPENDIX A where the response factor can be forced to give zero response for fundamental modes. With the varieties of designs, it is impossible to obtain complete cancellation for every resonant mode, but typically designs can be utilized with resonant factors down to at least 0.20, i.e., dynamic stress reduced by a factor of five, as compared to free-standing blades. However shrouding blades into packets also gives additional coupled modes to be concerned with as shown by Weaver and Prohl (1958) and many others, with actual strain gage data at speed in Heymann (1968).

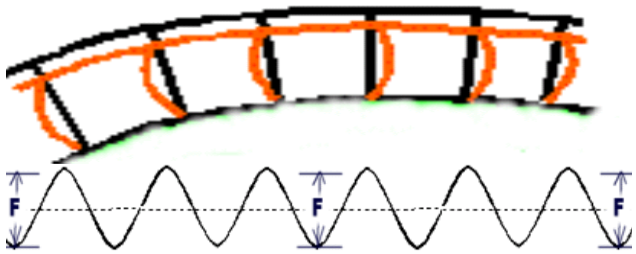


Figure 19. Out-of-phase mode; Some blades in-phase with excitation (F); Some blades out-of-phase with excitation

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By changing the number of nozzles, some blades in a packet are excited in-phase at the same time other blades are vibrating out-of-phase with exciting forces. Thus much of the excitation is cancelled and resonance with nozzle passing frequency can often be accommodated within the operating speed range. If the first out-of-phase mode (torsional packet mode) is the one resonant with nozzle passing frequency, nearly equal spacing could be a good design. Forces cancel out much to give a low resonant response factor. But one doesn't ever want equal numbers with equal spacing. An out-of-phase mode in resonance with upstream vane passing frequency could have the number of blades per packet optimized giving a different response factor and/or frequency.

Some manufacturers seem to standardize with a packet count such as five or six blades per packet, but it is sometimes very important to optimize a given stage. For example, a high pressure, high temperature Curtis stage with 64 rotating blades and 71 inlet vanes should have eight blades per packet ($R = 0.142$), not four per packet ($R = 0.728$) if a fundamental mode is resonant with 71 times speed. If it is not resonant, it could be acceptable to use four blades per packet. Besides shrouding shorter blades to reduce response at high frequencies from inlet vane passing frequency, "long-arc" blading can be used to minimize response of fundamental modes (refer to Ortolano, et al., 1981; and McCloskey, 2002). In a low-pressure turbine stage, instead of using 24 packets with four blades in each, a six-per-revolution resonance could have six packets with 16 blades per packet around the circumference. APPENDIX A can be used to check that the response factor would be zero when using h (excitation harmonic) equal to 6. If the shroud could not physically fit over the tangs during assembly but could fit with eight packets, the response can still have fairly low response.

Friction Damping Methods

Integral tip shrouding designs that just contact each other when assembled have potential issues in providing snubbing action when at speed especially with thermal growth transients. Besides shrouding, part-span snubbers and lashing wire designs have greatly reduced failures in steam turbines. The 1964 patent US3131461 for Zig-Zag pins expired; design is often used including by other manufacturers. Campbell diagram for one design is shown in Figure 20.

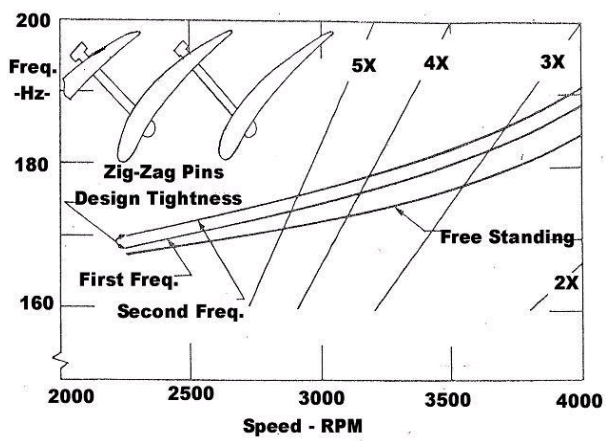

Figure 20. Rotating test results for zig-zag damping design.

The design has been reliable with proper operation but two other methods for friction damping have been used, one for decades: integral shrouds with rolled-in lashing wire and one more recently, continuous coupled Z-shrouds. One design configuration that has been effective, especially for shorter blades, and used for decades is the integral-shroud with one wire (Figure 21) and sometimes two rolled-in lashing wires on the top.

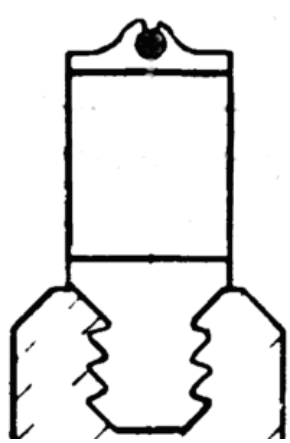

Figure 21. Rolled-in lashing wire (Year 1966 Vintage)

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In effect the design has many advantages with the integral shrouds acting as snubbers. They are assembled with zero or negligible gaps at the tips; the small gaps that occur at speed and temperature are usually small enough to limit motion especially in the flexible tangential direction. In addition, rolled-in lashing wire provides a large damping force where there is high relative motion between blade tips. This technique also utilized by other manufacturers offers friction damping without need to have the wires exposed in the gas stream. Some blades can have two rolled-in lashing wires as shown in Figure 22, with the ends of each wire meeting at blades that are 180 degrees circumferentially opposite.

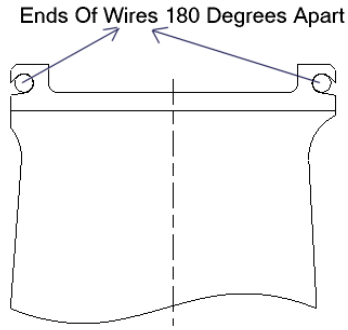

Figure 22. Rolled-in damping wire design with increased sealing surface

The trend today is to use many more Z-lock shrouds (Figure 23) for tapered, twisted blades, with friction damping at the tip contact surfaces, eliminating the need for lashing wire and/or shrouds with peened tangs. This design acts by having the blades "untwist" as the rotor comes up to speed further locking up at the tip. By having contact for all blades, the assembly vibrates with nodal diameter patterns. There are families of coupled modes where vibration has motion in the axial and tangential directions. The trend likely began with designs such as the power turbine for the gas turbine derivative. With a continuous shroud, the modes to consider are coupled modes where the blades vibrate with mode shapes defined by nodal diameters. Thus instead of individual blade modes, the most responsive disk-type critical speeds must be avoided as described later for disks. For example, the three-diameter mode of a design may have high stress if the frequency is equal to three times operating speed; for either of both sets of modes defined for motion in the predominant tangential and then axial direction.

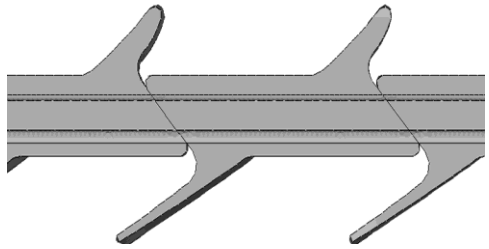

Figure 23. Z-Lock continuous 360-degree shroud

Figure 24 is an interference diagram for a z-lock shrouded design where the first mode could have resonance for the five or six-diameter mode for the first family of modes in the speed range. The second family of modes could have the nine- through 11-diameter mode in resonance. Thus there is need to limit high excitation sources to other than 5, 6, 9, 10, or 11 times speed. For some designs, the blades must be preloaded to ensure that under centrifugal growths and blade "untwisting", there is sufficient contact for the surfaces, which in turn are also properly coated. Other advantages of Z-lock shrouds are: permit ease of using tip seals; eliminate inherent aerodynamic losses of lashing wires and snubbers in the steam path; eliminate areas in open shrouds and lashing wire designs where deposits and corrosive media can accumulate.

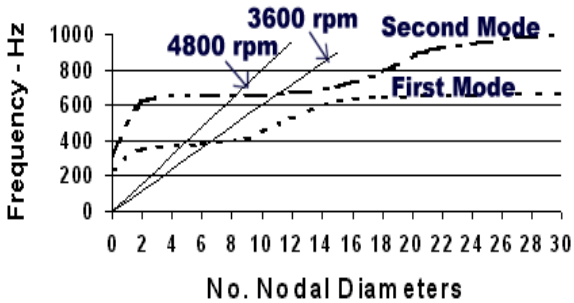

Figure 24. Z-lock design interference diagram – resonance above 4-diameter mode

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Proper and Improper Application of Phase Cancellation

Phase cancellation assumes that the blades are essentially in communication with each other, essentially being coupled together in some way. Thus, for blades in packets, it is possible to largely cancel out the motion of the overall packet because the blades are tied together by a shroud such that the motion of one blade affects the motion of the others. Individual blades within the packet can still be in a resonant situation for those modes where the blades are not in communication with each other. The application of phase cancellation between packets is certainly feasible where packets communicate through the disk rim; however, this ability quickly disappears when the rim becomes rigid compared to the packets. In particular, the rim is quite rigid in the tangential direction, having insignificant participation in the mode, and packets phase cancelling one another cannot generally be relied upon for tangential modes. In the axial direction, the rim is not so rigid, so phase cancellation between the packets is possible since the rim certainly participates in the deflection, but this participation becomes less and less as the blades are made more flexible in the axial direction usually due to blade height. In addition, mistuning can interfere with phase cancellation.

Turbine Blade and Packet Mistuning

As discussed, the problems associated with mistuning are as follows:

- Mistuning can potentially lead to large local vibration level around one or a limited number of blades, increase the stresses of blades compared to the perfectly tuned system, thereby increasing the likelihood of a high cycle fatigue failure
- Mistuning can interfere with phase cancellation if the level of mistuning encompasses adjacent modes.

For turbine disks with relatively short blades in packets, mistuning is often present such that the natural frequencies of each packet for a particular mode can be found a few percent different from packet to packet. The packets can be excited individually and the nodal diameter pattern may not lead to phase cancellation of the excitation force. In these cases, it is not recommended to only rely on interference diagrams as prescribed by others. Use of the diagrams gives an excellent method to initially select number of blades per packet, but phase cancellation equations also need to be checked to guard against mistuned modes. Mistuning of blades as described in many publications surely prevents the occurrence of such nodal diameters for many but not all cases, in that some blades can be largely decoupled from the disk, especially for the predominant tangential direction. Thus each packet can be excited separately; e.g., resonant with nozzle passing frequency; geometry patterns that define response are numbers of blades per packet and ratio of nozzle to rotating blade numbers for phase cancellation (refer to APPENDIX A). In fact, just as for "rouge" cantilevered blades without shrouds there will be increased response of some packets, in effect acting as vibration absorbers for others, (refer to analysis for shrouded blades in Bladh et al., 1999).

The reality is that there can be different frequencies for the same mode, including the fundamental, albeit close together; thus in some stages such as those with locking pieces instead of locking buckets, it is necessary to have different numbers of blades per packet that gives added differences. Actual strain gage data in operation is given in Heymann (1969), showing that all fixed-supported tangential packet modes responded at nozzle passing frequency, proving there was mistuning. If there ever was a troublesome mode of an entire structure, i.e. blades all coupled through the disk that they are attached to, the method of nodal diameters may have merit as has been well known.

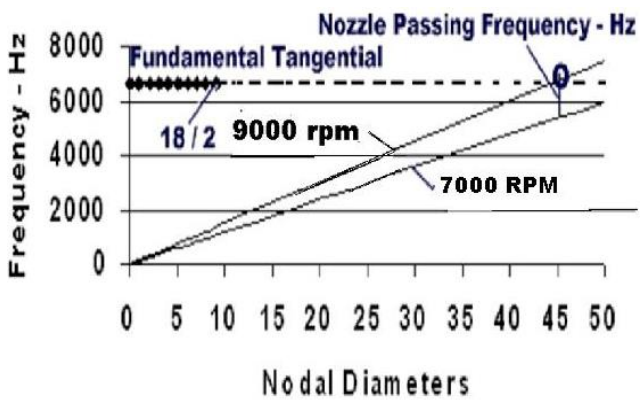

Figure 25. Interference diagram - 100 blades / 18 packets / 45 nozzles

Figure 25 is an interference diagram for a short turbine blade where the fundamental mode is above 6000 Hz. It is a design with a combination of five and six-bladed packets. The diagram assumes all packets have modal frequencies identical with all coupled together in diametral patterns. The shape of the nine-diameter mode or those below could only be excited as there would be nodal lines between each of the 18 packets, but "interference" avoids 45 times speed. The excitation for 45 inlet nozzles theoretically wouldn't excite the fundamental mode so it may be claimed that resonance wouldn't exist. However due to mistuning a particular packet can be excited - in this case at 45 times speed near 9000 rpm. The design likely is adequate as resonant response factor is below 0.20 that is a typical recommended maximum value (see Table 1 below).

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Table 1. Mistuned fundamental mode resonant response factors for 45x excitation; 100 blades, 45 nozzles

No. of Blades Per Packet	Resonant Response Factor
2	0.16
3	0.30
4	0.15
5	**0.14**
6	**0.14**
7	0.07
8	0.12
9	0.02

Figure 26 is for a revision with a change to 36 packets with three to four blades per packet - there would be a limit of 18 nodal diameters (36 divided by 2), so that 45 nozzles still do not match the modes. But again this is only true for modes where all packets vibrate with the same frequencies. In this case the 18-diameter mode would have 36 nodal lines. Higher diameter modes cannot get excited – similar to only needed to consider up to 18 diameters for response of disk modes with 36 cantilevered blades. In other words each packet is in effect treated as an individual blade so that only the 10-diameter mode would have interaction resonance. Since |45 nozzles – 36 packets| = 9 then a 9-diameter mode could be excited at 45 X speed, but the response would be low even if perfectly tuned. Also see Wagner and Griffin, (1996a, 1996b) who analyze packets with a different graphing method. Mistuned packets will negate this diagram completely and the resonant response factor would be 0.30 as found with phase cancellation equations, see Table 1. Thus an individual packet could have excessive response with a response factor well above 0.20.

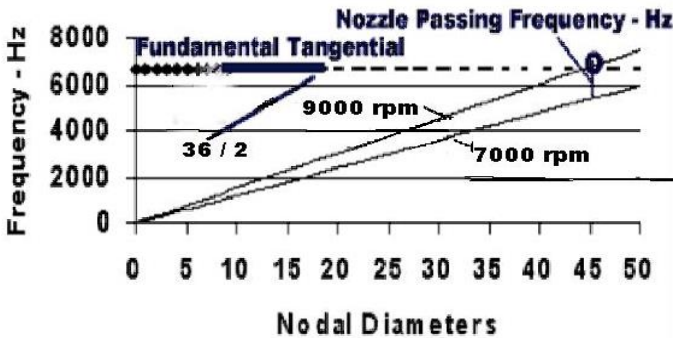

Figure 26. Interference diagram for 100 blades / 36 packets

Modes with amplitudes out-of-plane of the disk could more easily have coupled modes. To verify interference diagrams it is suggested to double-check for acceptable response factors with a proven method, phase cancellation equations as explained by Weaver and Prohl (1958). With long tapered twisted blades with relatively flexible disks, the graphing method is useful for coupled bladed disk-dominating modes. Also as explained by Wagner and Griffin, (1996a, 1996b), it is the modes that have out-of-plane motion that are effectively coupled. For designs with very rigid disks or drum construction, individual packets of short, non-twisted blades, phase cancellation analysis is the method to use, especially for the tangential modes.

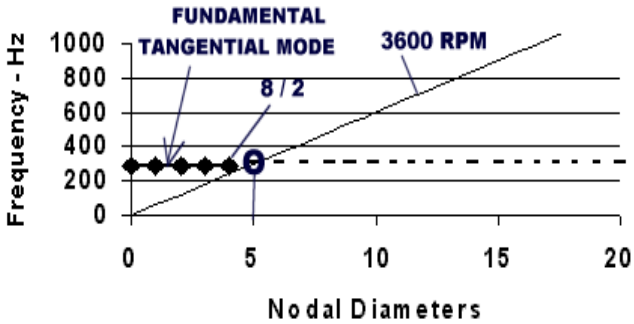

Figure 27. Interference diagram; 100 blades / 8 packets

Another example in Figure 27 shows a change in packet size for a steam turbine long blade design. The original design had 18 packets with interference at 5 times speed when at 3600 rpm. However by changing from 18 packets (five and six blades per packet), to less than nine packets, there would not be an interference point. For example, if there were eight packets, the point shown in Figure 27 where (8 / 2 = 4) would be less than 5 times running speed. If all packets were indeed tuned, Wagner and Griffin (1996a, 1996b) show that the

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3-diameter mode could still be excited somewhat if equal to five times running speed. In this case, if packets were mistuned, recommending a change from 18 packets to 8 packets will reduce the mistuned resonant response factor from 0.86 to less than 0.50; see Table 2. Also since there are a different number of blades for packets in the assembly, that automatically gives some mistuning. As compared to short blades, longer blades have small effects of root / disk rim contact differences blade-to-blade on modal frequencies. Response factors less than 0.5 is typically acceptable for low harmonic excitation such as five times speed.

Table 2. Mistuned fundamental mode resonant response factors for 5x excitation; 100 blades.

No. of Blades Per Packet	Resonant Response Factor
6	0.86
12	0.51
13	0.44
14	0.37
15	0.30
20	0.00

Eight packets (12 to 14 blades per packet) would usually be safe especially if the stage is a not a transition stage with wet steam and potential corrodants in the steam. The original design with six blades per packet could have too high of a resonant response factor near 0.86 for mistuned packets. However by using the "long arc" method previously described, if the Goodman factor is very high, it would be optimum to have five groups of blades tied together, 20 blades per packet giving a resonant response factor near zero. This is called the "long arc" method of selecting blade packets for steam turbine back-end stages. Standard designs with as many blades per packet as can easily fit the tangs are generally much better than those with low numbers such as four to six blades per packet.

Impeller Design and Resonance Avoidance/Mitigation

The first check that should be made is to avoid disk critical speeds. As already discussed, this is typically not a problem for covered impellers, but open impellers can be prone to having a disk critical speed, especially if made from titanium or other lightweight material that allows for extremely high speeds. For covered impellers, disk critical speeds are typically below the operating speed requiring a traverse at low speeds. Traversal of disk critical speeds has not been a problem to date. Steady-state stress is low and reduced power provides lower exciting forces that further limits the risk.

The second check is with respect to vane passing excitation. There are two stationary sources of concern: (i) upstream vanes such as inlet guide vanes, sideload guide vanes, and return channel vanes; and (ii) downstream diffuser vanes. The upstream vanes excite blade leading edge modes and disk modes whereas the downstream vanes excite blade trailing edge modes and disk modes. The distance from the stationary vanes to the rotating blades is of some importance here. The greater the distance, the less likely it is for wakes to cause impeller excitation. Diffuser vanes are a particular concern as they are typically located close to the impeller tip. Excitation is also known to increase when approaching surge and also at overload approaching choke flow (see case B-2 below). Interaction resonance of disk modes based on differences between rotating and stationary blade numbers also need checked using equations or interference diagrams.

Open impellers are capable of faster operation than covered impellers. The additional speed will naturally lead to higher frequencies of excitation. These impellers have lower disk mode frequencies than their covered counterparts. In addition, open impellers have individual cantilevered blade modes much lower than designs where a cover gives fixed-fixed boundaries. Freestanding, cantilevered blades used for open-impellers also have much more risk for HCF than fixed-fixed blades in covered impellers. They are especially prone to fail with excessive compressor surging, as are axial compressor blades. The most responsive blade modes are designed to avoid excitation from inlet guide vanes – fixed or adjustable. If there is minimal mistuning, blade modes can be coupled to each other in diametral patterns. Again a caution is always to avoid equal blade and vane numbers, checking inlet and diffuser vanes. For impellers with splitter blades, both full or splitter blade numbers and their sums should be evaluated for excitation.

As impeller sizes increase, and in particular the throat area, the leading edge flapping mode can become a design issue for covered impellers that should be addressed. The fixed-fixed mode is not nearly as compliant as a cantilevered blade, but if the blade is long enough, blade sculpting where the hub side of the blade is thickened may need to be employed to avoid a resonant situation (Figure 28).

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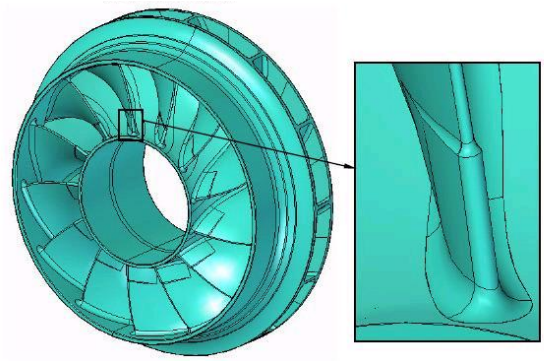

Figure 28. Sculpted blade example

Schiffer and Syed (2006) reported on two blade leading edge failures that were caused by resonance, requiring a change from 16 to 12 inlet guide vanes. Cases such as described by Singh, et al. (2003), and Phillips, et al. (2003), for two times number of inlet vane excitation could be due to other mitigating factors, and certainly raise questions such as:

- Was operation within the specified operating map?
- How much liquid was really being ingested to potentially aggravate the wakes from upstream vanes and reduce aerodynamic damping?

Impeller Damping

As compared to turbine blades, impellers do not have damping treatment options. The damping that does exist is from the material and from aerodynamic damping. Aerodynamic damping is expected to provide the most damping. There have been some suggestions that coatings can also provide damping, but this is generally considered unreliable as many coatings wear away over time.

Determining Natural Frequencies and Mode Shapes

In order to avoid natural frequencies, it is first required to know where the natural frequencies are located. The primary method for determining natural frequencies used to be testing. Today, natural frequencies are mainly determined by finite element analysis with testing taking on the role of verification when needed. For modal analysis using the finite element method careful modeling is required (Figure 29) with proper techniques including proper boundary conditions. In order to obtain relative stresses for higher order modes, the entire impeller may require modeling.

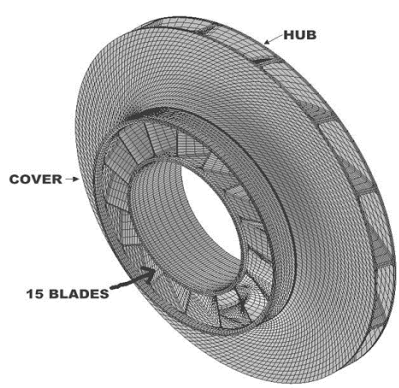

Figure 29. FEA model of covered impeller with full inducer and curved, 3-D blades (Kushner, et al., 2000).

Modal analysis of a cyclic symmetric structure to determine nodal diameter frequencies can be computed by modeling only one sector per the finite element computer code (Kohnke, 1999). Constraint relationships can be defined to relate the displacements of one edge of the cut boundary to the other edge. This allows for calculation of natural frequencies related to a given number of nodal diameters. The basic sector is used twice to satisfy the required constraint relationships and to obtain nodal displacements. The analysis results will show pairs of frequencies for each nodal diameter solution. Using this technique, it is possible to obtain solutions for up to a number of nodal diameters, n, up to n = B/2. For an even number of blades, B, this would be B/2; for an odd number of blades this would be (B-1)/2.

As an example, a 13 blade impeller will have a sector equal to 360 divided by 13 which is a 27.69 degree section of the circular impeller (Figure 30). An additional advantage of this cycle symmetry is that it classifies the modes based on nodal patterns. Welds are modeled explicitly where the blade and hub and cover join. Bonded contacts are used to simulate the welds and frictionless contacts are used between the blade and hub/cover unwelded interface. The FEA is modelled with the impeller installed on the rotor. In general, the sector model for this impeller had roughly 125,000 elements and 200,000 nodes. The mesh size was chosen based on a mesh sensitivity study

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and variation in frequency was less than 0.2%. This sector model can determine the blade and nodal diameter natural frequencies and modes of the entire disk.

Comparing Figure 29 (Kushner, et al., 2000) to Figure 30 (Pettinato, et al., 2024) shows the evolutionary advance of the computational model after roughly twenty-four years. The mesh is finer and the tools are more sophisticated.

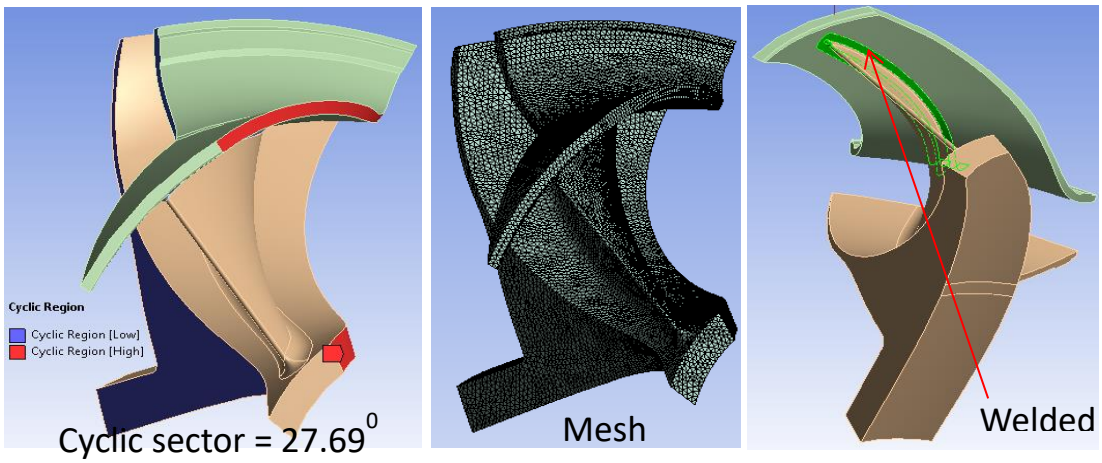

Figure 30. FEA impeller sector modelling (Pettinato, et al., 2024)

Modal Testing and Natural Frequency Accuracy

A test campaign was conducted where blade leading edge modal test data were collected for a wide range of centrifugal impeller sizes and compared to the finite element analysis predicted modes (Pettinato, et al., 2024). Modal tests were performed using the roaming hammer method. A single stationary fixed accelerometer is attached to the impeller and force is applied successively at different locations with an instrumented hammer (Figure 31).

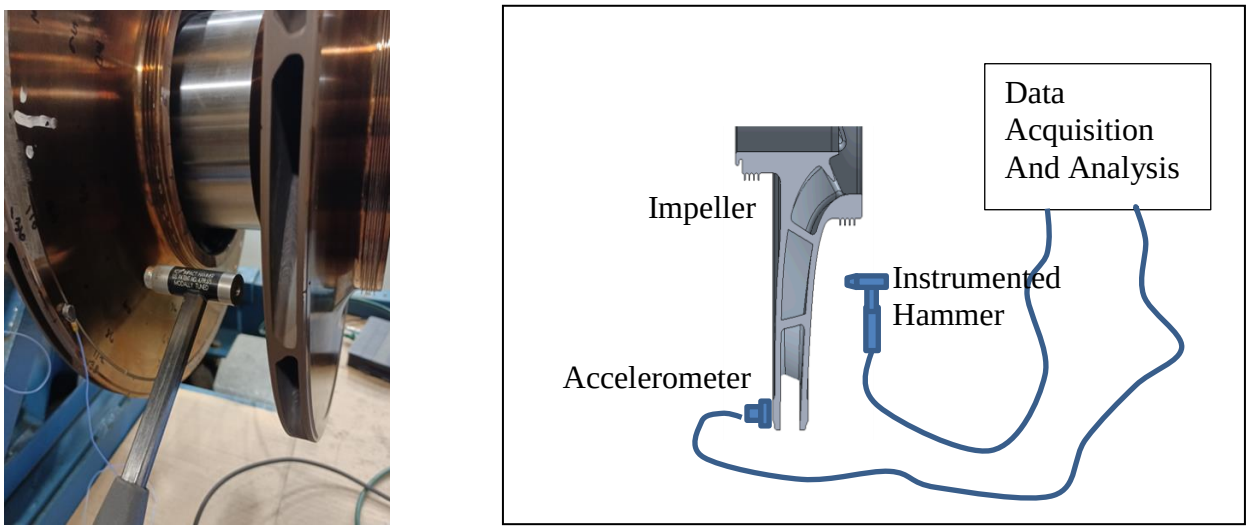

Figure 31. Modal testing by roving hammer method

Figure 32 shows the predicted and measured 2-diameter disk modes. To better illustrate the mode, the quarter section marked with the + symbols can be considered as moving out-of-phase with the – symbols.

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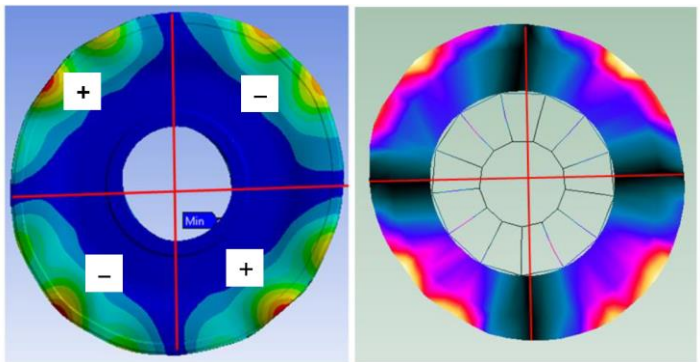

FEA ND2 MODE ANIMATION MEASURED ND2 MODE ANIMATION

Figure 32. Impeller ND2 mode shapes

Figure 33 shows the results for one of the blade leading edge modes after testing over thirty different impellers. The 0% datum is the FEA prediction as it is essentially without variance. 0σ represents the mean of the test data. The data for this specific mode is skewed higher than prediction. The analysis of the data resulted in a reduction and shift of the original uncertainty bands and also more confidence in the application of the FEA predicted models.

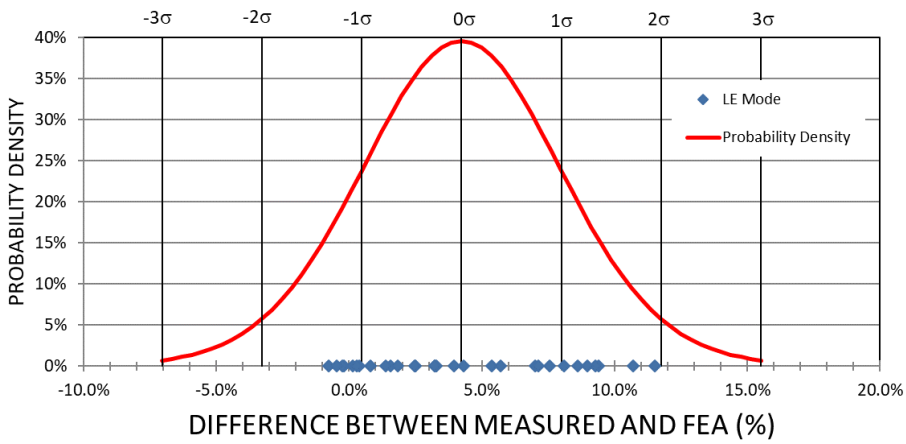

Figure 33. Blade leading edge results for thirty impellers and one mode

Figure 34 shows the level of variance for three identical impellers.

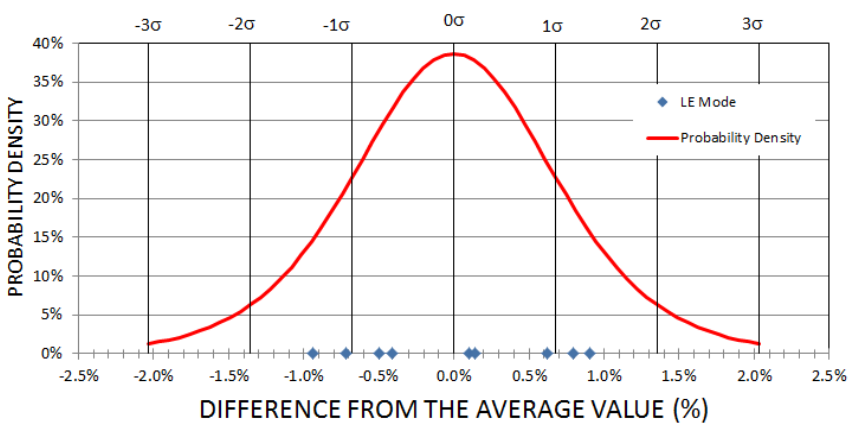

Figure 34. Repeatability plot of three identical impellers

OPERATING IN RESONANCE

When operating in resonance, the vibration level is only limited by damping, some other limiting mechanism, or breakage. However, successful and intentional operation in resonance occurs quite often. Continuous operation in resonance is typically approached by performing a fatigue analysis. Such analyses are fairly routine for turbine blades. Fatigue analysis can also be performed on centrifugal

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compressor stages though the cost and time typically restricts this analysis to root cause failure investigations. The key to operation in resonance is to keep the vibratory stress below the endurance limit. The vibratory stress was earlier defined.

$$\sigma_v = \sigma_s \cdot S \cdot R \cdot \left(\frac{\pi}{\delta}\right)$$

The steady bending stress, σ_s, is determined by the stage power. The stimulus, S, also known as unsteady loading can occur due to various sources of excitation as will be discussed later and is typically determined by performing a fluid structure interaction analysis (FSI) utilizing computational fluid dynamics (CFD). The resonant response factor, R, is determined by packeting and nozzle count, and is typically determined by a phase analysis similar to what is provided in Appendix A. The amplification factor, π/δ, is dependent on damping. The vibratory stress σ_v can then be determined by solving the given equation or it can be determined directly by a forced response analysis involving the structure.

Steady Bending Stress

The steady bending stress is due to the aerodynamic load. For blades, the key stress points include the base of vane and the roots. For impellers, the key stress points include the blade leading edge, the blade trailing edge and cover tip.

Unsteady Loading

Unsteady loading in turbomachinery blades and bladed disks can occur due to various sources of excitation. Excitation can come from non-uniform flow and pressure due to construction geometry besides the inherent wake effects from adjacent vanes. An example would be harmonic excitation at multiples of speed from poorly designed inlets or exhausts. The force distribution can be separated into pulse trains for the analysis. As speeds increase, shock waves become a consideration for normal operation, besides having potential for transient conditions or those inadvertently outside operating performance maps. Differences in manufacturing such as variations in upstream throat areas between vanes can give significant excitation, as can geometric alteration over time from erosion and corrosion differences circumferentially. For compressors, aerodynamic sources also can include rotating stall and surge at one end of the operation map and choke flow on the other. Turbines are inherently more stable, but also more prone to have changes to both excitation forces and endurance strength due to erosion, corrosion, and deposits. In addition, partial admission stages give a rich signature for many harmonics of speed. In effect, as a blade enters the arc it can be in-phase, vibrate a number of cycles, and then enter the arc in-phase for the next revolution, i.e., in exact harmonic resonance. The higher the number of cycles of vibration per revolution, the lower the response as there are more cycles having a decay of amplitude via damping mechanisms. Partial admission can also aggravate the excitation from the worst offender, wake-passing frequency. The predominant source of excitation for short blades is passing frequency, number of upstream wakes per revolution. Downstream wakes are usually of little concern, along with most cases of difference frequencies between upstream and downstream vanes. Passing frequency is based on the spacing between wakes; Fourier analysis can be utilized for non-uniform spacing of nozzles and vanes.

Sources of unsteady loading are listed by API 612 (Steam Turbines) as

- first and second passing frequencies of upstream and downstream stationary blade rows;
- steam passage splitters;
- irregularities in vane pitch at horizontal casing flanges;
- the first four turbine speed harmonics;
- casing openings (exhaust or extraction);
- partial arc diaphragms or nozzle plates;
- internal struts and structural members in the inlet and exhaust casing or horizontal joints.

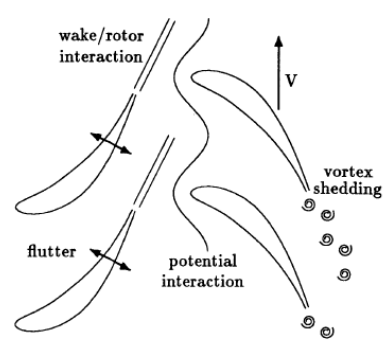

Figure 35: Source of unsteadiness in turbomachinery blade rows [Giles]

Additional mechanisms such as vortex shedding, acoustic coupling, flutter, and shock wave interaction are generally more important for other turbomachinery such as gas turbines and compressors. A brief explanation of some excitation mechanisms is provided:

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Wake/Rotor interaction:
Wakes from upstream stator which can be considered as approximately steady in the stator frame of reference are unsteady in the rotor frame of reference as the rotor blades are moving though the wakes causing unsteady forces on the rotor blades. This is typically primary source of stator/rotor excitation. These can be due to upstream nozzles in steam turbines; stator aerofoils in axial turbomachinery; or stator guide vanes in radial turbomachines such as inlet guide vanes; return channel vanes to guide flow between compressor stages; or side load guide vanes allowing flow from sidestreams in a centrifugal compressor.

Potential interaction:
Pressure in the region between the stator and rotor blade rows can be decomposed approximately into a part that is steady and uniform, a part that is non-uniform but steady in the rotor frame (due to lift on the rotor blades) and a part that is non-uniform but steady in the stator frame (due to the lift on the stator blades). As the rotor blades move, the stator trailing edge experiences an unsteady pressure due to non-uniform part that is locked to the rotors, and the rotor leading edges experience an unsteady pressure due to non-uniform part that is locked to the stators. This is a purely inviscid interaction which is why this is labelled as "potential" interaction. This type of interaction is particularly important in cases where stator-rotor vane spacing is typically smaller such as in axial turbomachinery as well as impeller-diffuser vanes in radial turbomachines.

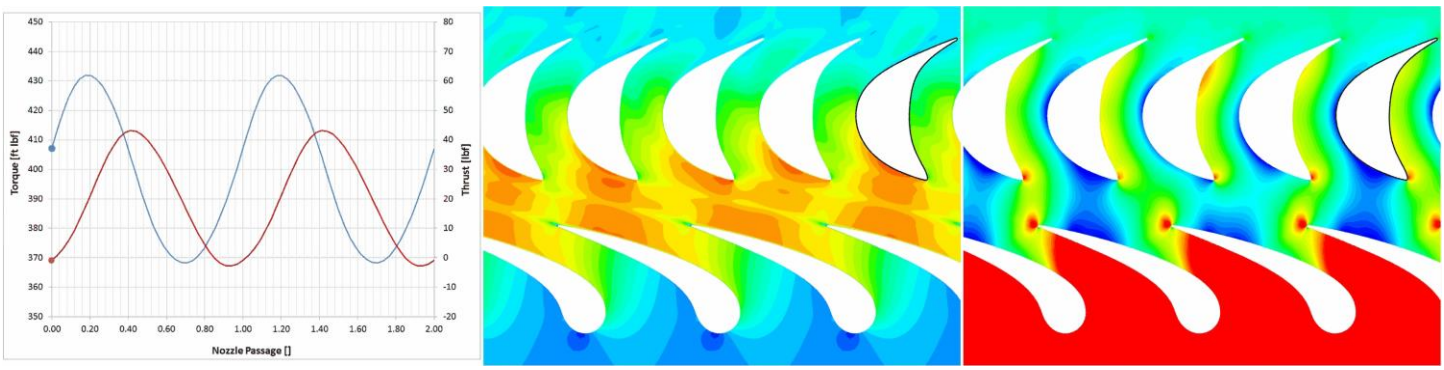

Figure 36. Steam turbines experience both wake interaction and potential interaction

Inlet and Discharge Distortion:
Non-axisymmetric flow conditions at the inlet due to bends (such as inlet nozzles, side stream inlet nozzles in centrifugal compressors) or partial arc admission in steam turbines or the discharge of axial compressors and steam turbines can also create unsteady loading. The excitation frequencies and magnitudes due to inlet distortion can be analyzed by Fourier series decomposition of the circumferential flow variation.

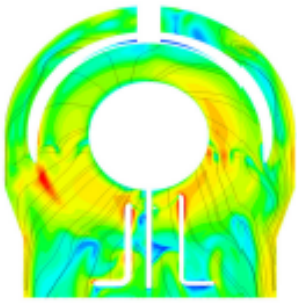

Figure 37. Flow Asymmetry in the Exhaust Hood Flow Structure

Acoustic Interaction:
Acoustic interaction is discussed in detail in later section of this tutorial. Acoustic interaction can arise from interaction due to Tyler-Sofrin acoustic modes (Konig, et al., 2009), acoustic mode interaction in cavity flows with the rotating and vibrating bladed disk. Acoustic modes have the potential to influence aero-dynamic damping.

Flutter:
Motion of stator/or rotor blades may also introduce unsteadiness. Flutter is a condition in which oscillation of the blade may produce unsteady forces and moments on the blade. These unsteady forces/moments due to the phase relation of the motion may impart energy to the blades increasing the amplitude of blade's unsteady motion. This case is also referred to as negative aerodynamic damping. If energy is extracted from the blades and causes a reduction in amplitudes, the system is said to have positive aerodynamic damping. For analysis of excitation methods to actually calculate response requires FSI analysis methods that combine CFD and FEA. Care must be used to separate modes and respective damping. A direct approach is given by Petrov (2009) and Allison et al. (2011).

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Wake-Rotor/Potential Interaction in Centrifugal Compressors:
Wake-rotor/potential interaction in centrifugal compressors arises from inlet guide vanes, side-stream inlet guide vanes, return channel vanes and diffuser vanes. Figure shows the components on a centrifugal compressor cross-section. The frequency of excitation on rotating components from these stationary sources is typically the stationary vane passing frequencies. However, non-uniformity of flow characteristics or inlet distortion (especially inlet/side stream inlet bends) or non-axisymmetric vane spacing especially in inlet/side stream inlet vanes can contribute towards additional harmonics of running speed besides the vane passing frequency.

Forced response analysis including High Cycle Fatigue analysis using High Fidelity CFD and Finite Element Analysis
HCF analysis of centrifugal compressor impellers require use of high fidelity methods as the simplifications such as beam model equivalents in axial turbomachinery blading is not possible. This necessitates the use of structural analysis of the impeller using techniques such as Finite Element Method. For a tuned system, modeling of only one sector is possible by taking advantage of the cyclic symmetry boundary conditions and reduce computational time for structural analysis. Steady state stresses due to centrifugal effects and aerodynamic gas loading can be estimated using FEM. Aerodynamic gas loading requires the estimation of steady blade pressures from a CFD analysis or can be estimated using a meanline compressor efficiency analysis tool. Unsteady stresses on impeller blades depend on the unsteady force magnitude acting on the blades due to various sources of excitation, unsteady force phase in relation to the blade vibratory motion and damping in the system. Typical sources of excitation such as wakes from IGV, return channel systems, etc. are traveling wave types of excitation and the response is also of traveling wave in nature. Estimation of maximum stresses, therefore, requires a phase-sweep analysis for such kind of excitation. Unsteady aerodynamic pressures on impeller blades can be obtained from CFD analysis or can be estimated using simplification such as percentage of steady aerodynamic loading obtained from mean-line analysis. The simplification, however, needs to be correlated with high fidelity CFD analysis or experimentation.

The primary source of damping in centrifugal compressor impellers is aerodynamic damping. Material damping is typically a small fraction as compared to aerodynamic damping but can be important especially in low gas density applications. Aerodynamic damping can be evaluated using high fidelity CFD analysis or can be estimated using experimental means. The overall process for estimating steady and alternating stresses on a typical centrifugal compressor impeller is explained in further detail below:

Alternating Stress Estimation
Alternating stress can arise from various sources of excitation as discussed earlier. The general equation of motion for aerodynamic excitations in an impeller is represented by:

$$[M]\cdot\{\ddot{u}\}+[C]\cdot\{\dot{u}\}+[K]\cdot\{u\}=\{F\}_{harmonic}+\{F\}_{aero}+\{F\}_{unsteady_general}$$

Here, u represents the displacement degree of freedom for the nodes of the finite element mesh. $[M]$ is the mass matrix, $[C]$ is the damping matrix, and $[K]$ is the stiffness matrix.

$\{F\}_{harmonic}$ *=Forces due to harmonic excitation due rotor-stator interaction*
$\{F\}_{aero}$ *= fluid structure interaction forces*
$\{F\}_{unsteady_general}$ *= general unsteady forces due to non-uniformity of flow, turbulence, etc.*

Aerodynamic mass and stiffness influence the natural frequencies of the structure and damping typically influences the response magnitude. For typical centrifugal compressor applications with low and medium pressures and gas densities, the influence of aerodynamic mass and stiffness can be considered to be negligible. Aerodynamic damping is also of the order (generally < 1% modal damping for typical applications) such that the structure can be considered lightly damped. Hoopes, et al., (2020) discuss the estimation of aerodynamic damping in a centrifugal compressor impeller. The typical work-flow in the analysis includes a modal analysis to determine the structural modes and frequencies of interest for the impeller. The motion of the mode is then transferred to the CFD analysis (URANS) as a deforming mesh boundary condition to estimate the aerodynamic work on the impeller surfaces. This aerodynamic work done per cycle of the mode frequency is then used to determine the aerodynamic damping for the given mode. This modal damping is then used as an input in the structural harmonic analysis to estimate the alternating stresses.

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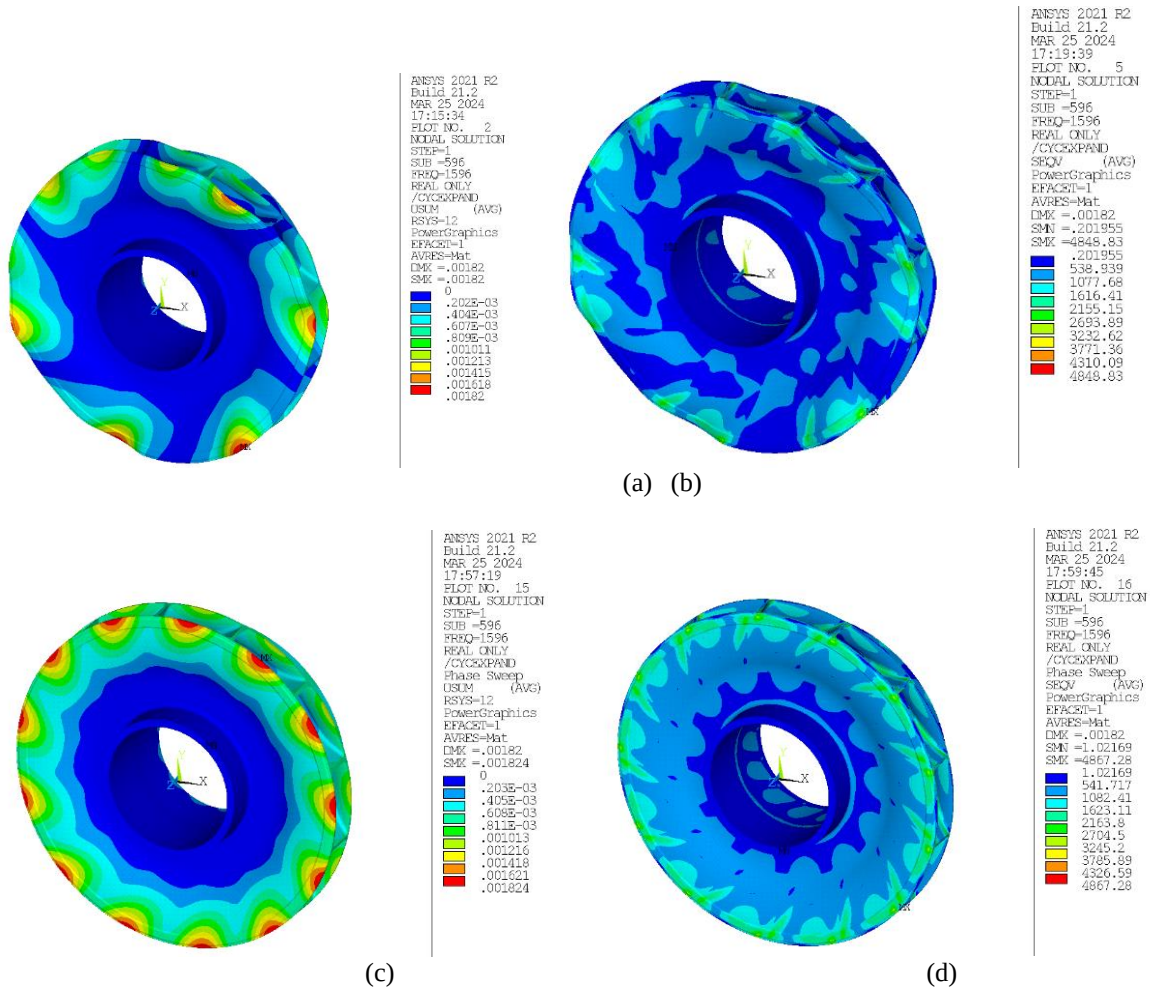

(a) (b)

(c) (d)

Figure 38: Response of an impeller with 13 blades to 16 Engine Order Aerodynamic excitation

Fatigue Evaluation

The estimated steady and alternating stresses is further used in fatigue analysis. For centrifugal compressor impellers, the problem of fatigue is of the nature of high cycle fatigue as the number of stress cycles due to typical excitation sources can reach > 10^6 cycles within a short period of operation. Stress life method is typically considered for high cycle fatigue analysis with the results of alternating and mean stresses plotted on Goodman-Haigh diagram. The limits of operation on the Goodman diagram are based on material endurance limit for the alternating stress and the tensile strength for the mean stress. Appropriate fatigue strength modifiers and safety factors are required to estimate the design limits. Evaluation of welded connection requires special treatment as discussed by Hoopes, Leavitt, et al., (2020). Several international codes are available for fatigue evaluation such as ASME BPVC Section VIII, British standard BS 7608, European standard EUROCODE3, and others.

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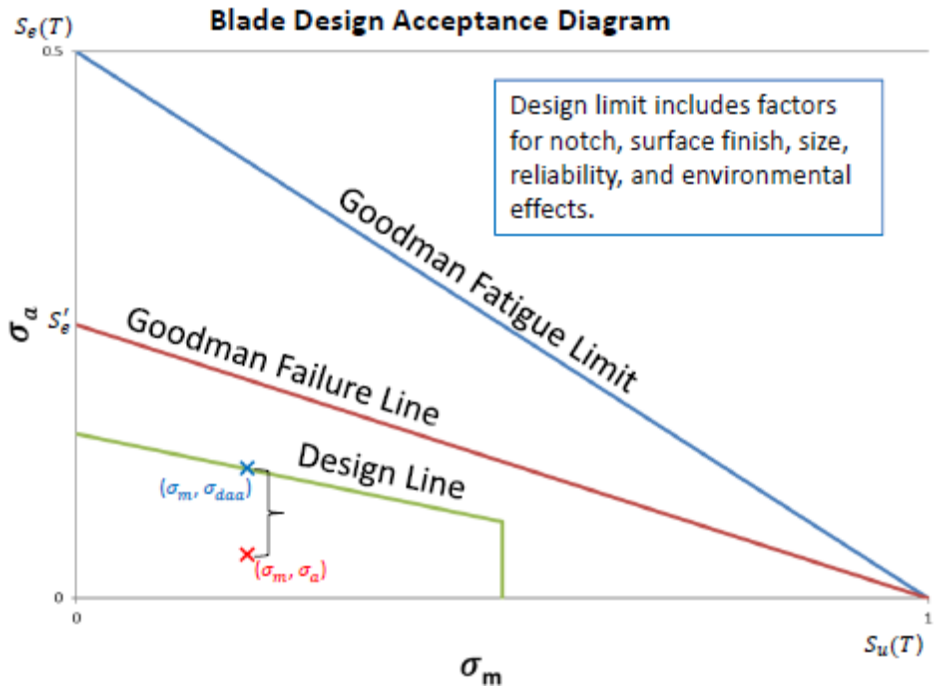

Figure 39. Blade Goodman diagram

High Cycle Fatigue

Failures will occur when the vibratory stress exceeds the endurance limit either through improper design/manufacture, improper operation, or because of degradation such as material corrosion. Most fatigue failures that are encountered are diagnosed as being high cycle fatigue (HCF). HCF failures result in cracking or fracture from a large number of stress cycles below the yield strength of the material and is associated with fatigue lives greater than about 10,000 cycles. Low cycle fatigue (LCF) failures involves higher forces that approach and exceed the yield strength of the material and is associated with fatigue lives less that about 1,000 cycles. The transition from low cycle to high cycle fatigue generally occurs between 1,000 and 10,000 cycles. For design purposes, the transition is cited as 1,000 cycles as it is conservative. LCF is normally not an issue for blades and disks in machinery to be covered herein though it must be noted that severe forces from improper operation such as continuous surging and/or excessive liquid ingestion can cause failure in much less than a million vibration cycles.

For many observers, even 10 million cycles (corresponding to threshold endurance strength for many defect-free, ductile materials) appear to be an unattainable high number of cycles for setting a lower limit. However, consider that structures discussed in this tutorial respond in the range of 200 to 10,000 cycles per second. Depending on the severity of excitation and resonant amplification for the structure, a failure can occur in a few minutes to several hours if operating at high loads very close to, let alone exactly at, a resonant operating speed. There have been cases of failures in the first day of operation after startup, as shown by one of the case studies.

Resonant response is avoided or minimized by controlling the stimulus or by changing the responding frequency. The vibratory stress can be controlled by changing the frequency of excitation, changing the strength of the stimulus or its relative phase with respect to the blade or bladed disk. The most common method of modifying blade frequency is to tailor the thickness of the blade or disk along with other variables affecting frequency: aspect ratio, taper, number and thickness of blades, and radius ratio of the disk outer to inner diameters. More drastic modifications are to use one of many damping methods, e.g., change to a friction damping mechanism at part span or by using a tip shroud incorporating damping and/or phase cancellation.

Operating changes can also cause HCF failures. Forcing can change and becomes excessive, such as from upstream or downstream blade or vane damage, operation at off-design causing rotating stall, surge, flutter, or other damage, e.g., from blade tip rubs. In some cases there can be simultaneous sources of excitation affecting the same or different modes. Corrosion, wear, and erosion can reduce resonant frequency margins and/or endurance strength over time. Either or a combination of these conditions can result in high-cycle fatigue initiation and propagation.

Once a structure has cracked, the critical stress intensity becomes lower such that crack growth requires less stress and fewer cycles as compared to crack initiation. If inspections do not find cracks to abate further propagation, crack growth then could easily eventually reach its fast fracture limit. Many machines have been shut down due to changes in monitored vibration data before a catastrophic wreck. Thankfully ductile materials help to mitigate this concern. After crack initiation, resonant frequencies will shift thereby either enhancing or reducing the vibratory stress. This change could give some relief, especially for constant-speed machines such as turbine/generator

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sets. In other cases, sometimes after a long period, resonance and/or the same or another speed change or off-design upset giving high loads can reoccur, causing the "beach-mark" patterns (showing start /stop events on a fractured surface).

Damping and Other Excitation Considerations

For blade flutter avoidance of long cantilevered blades in axial compressors and steam turbines: Σ aerodynamic and mechanical damping must be > zero. Empirical factors are often utilized for reduced frequency factors, in effect limiting how flexible the blades can be, e.g. limiting the fundamental modes to a minimum value for different machinery types. In addition, operation must have guidelines such as minimal surge and choke operation for axial compressors, and exhaust pressure boundaries for steam turbine exhaust ends. Precise analytical methods incorporating computational fluid dynamics (CFD) methods are still in development; in fact, especially for transonic stages and for two-phase flow, there are issues relative to inclusion of complete viscous and turbulence models even for steady-state conditions. Flutter aeroelastic calculations depend on the solution of unsteady nonlinear fluid mechanics, a Navier-Stokes solver, a solver for eigenvalues with structural equations, as well as a transient structural dynamic solver. Thus both dynamic mesh algorithms for the fluid mechanics and structural interaction are required. Nowinski and Panovsky (2000) and Panovsky and Kielb (2001) provide analysis and experimental data for low-pressure turbine stages for jet engines that once again verify that mistuning of blades greatly assists in preventing flutter. However, mistuning used to assist in guarding against flutter such as in steam turbine fixed-speed units will change the separation margins for any tuning for resonance avoidance. The authors also show how movement of the torsional axis from forward to aft position of the profile can greatly increase the propensity of flutter for the torsional mode.

Sources of blade damping are:

- Aerodynamic (viscous, normally positive but can become negative at poor incidence angles).
- Friction (lashing wire, mid-vane snubbers, Z-lock contacts, platform dampers, root/disk land interface.
- Hysteretic (material)—significant for many of the 12 percent chromium, martensitic steel alloys, small for most other steel alloys, and negligible for titanium.

Total system damping is one of the most difficult aspects of Equation (3). Values in Table 3 from Kielb (2001) are for materials with very low material damping and show a wide disparity.

Table 3. Amplification Factors from Jet Engine Strain Gage Data

Mode	Total Q
Fundamental Bending	30 - 130
Second and Third Bending	80 -100
First and Second Torsional	20 - 190
Stripe (Plate-type) Modes	60 - 150

Stationary components have a much lower failure rate as compared to rotating components. All cases presented herein involve rotating component failures and this is to be expected. Stationary vanes only have gas loads and potential thermal effects to give steady-state stresses. Without centrifugal loads, there typically is much lower mean stress, thus higher damping values as compared to rotating blades. Many vanes, especially those in steam turbines and the high-pressure sections of industrial axial compressors are also shrouded, giving reduced compliance with forced excitation as well as having inherent phase cancellation. Resonance of stationary vanes is much less likely but can occur and must sometimes be scrutinized, such as:

- Avoidance of resonance with rotor blade passing frequency of cantilevered diffuser vanes immediately downstream of rotating blades in a centrifugal design, and
- Rotating stall with multiple stall cells in compressors.

Acoustic Excitation and Response

One of the least understood causes of disk vibration has been acoustic interaction with acoustic pressure waves that directly excite the disk. The interaction of pressure waves between disk surfaces and the adjacent stationary wall may also be important, as shown by Eckert (1999) and Ni (1999); the instability of the two-diameter mode from acoustic interaction described could thus also give forced excitation with matching phase angles with the disk modes. It is rare whenever acoustic pressure pulsations reach high enough levels to cause turbomachinery components to fail. For a given sound pressure level such as 150 dB inside a compressor near atmospheric pressure, pulsations traveling at sonic speed are only approx. 0.1 psi (rms). Thus there also has to be an associated acoustic mode of the gas to increase excitation, such as can occur with compressor blade passing frequency in attached discharge piping, e.g. Kushner et al. (2002). But there is some confusion in that some investigators have said to also avoid direct resonance where rotating blade passing frequency is equal to a compressor disk mode frequency.

In the 1960's there began much insight into jet engine blade passing frequency noise by Tyler & Sofrin (1962), a widely referenced paper. They showed that tonal noise has inherent number of circumferential lobes based on phase analysis of numbers of rotating blades and stationary vanes; acoustic waves spin with modal patterns equal to the sum and differences of rotating and stationary blade numbers, including harmonics that can also be important.

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Besides this interaction there is concern for vortex shedding exciting acoustic modes within the annulus that contains axial compressor cantilevered blades. A case of blade failures was analyzed for excitation due to vortex shedding from inlet struts. It was shown not to be the cause but cracks were initiated by excessive surging of the system with an undersized blowoff valve. (See Case A-2). However treatment of the inlet vane shapes greatly reduced noise. Jet engine designers must guard against acoustic resonance for cantilevered blades.

Turning now to impellers for centrifugal gas compressors, there has been some concern that past fatigue failures had acoustic pressure pulsations as the excitation source. Eisinger and Sullivan (2006) described a case due to excessive turbulence at the inlet of a centrifugal fan. Price & Smith (1999) showed that noise caused by an inlet valve was eliminated, as it was the acoustic source for a covered impeller failure, albeit it may have been a different mode than what was described. Another source of high noise within machinery is flow past casing cavities, exciting modes within the cavity, but would be an extremely rare cause of structural fatigue for rotor components.

Normally the highest noise source in compressors is found to be at rotating blade-passing frequency. The wavelengths are typically small enough to have little attenuation as the sound travels up and down piping, thus exciting acoustic modes within the pipes. Resonance can be a concern for piping walls with matching structural modes, but even more commonly have caused cracks at protruding take-offs for instrumentation, etc.

The question under consideration is whether rotating blade passing frequency noise can actually cause self-excitation of impellers. It appears so but only in rare cases and likely mainly for high-pressure compressors where noise levels can be much higher. Richards et al. (2010) utilized strain gauges to confirm disk mode acoustic resonance. This report did not show a resonance at the rotating blade passing frequency but rather found the resonance was due to a combination of upstream vane wakes and downstream vane acoustic reflections. In their case there was not a corresponding acoustic resonance of the gas that was coincident with Tyler / Sofrin modes, so that dynamic strains were not of concern.

Two other recent papers brought the potential of acoustic interaction with impeller structural modes to full understanding with conclusive proof. Petry et al. (2010) confirmed Root Cause Failure Analysis given in the paper by Konig (2009). The two papers give analytical and confirming test results for two actual high-pressure compressor cases where modes of failed impellers were resonant at harmonics of stationary vanes – not at rotating blade-passing frequency. It was concluded there was resonance for 17 bladed impellers, 5-diameter disk mode with resonance at 22 times speed from downstream vanes, along with side cavity acoustic 5-diameter mode at 17 per rev resonance deemed to cause a failure. The fix for both cases was to change number of stationary vanes, from 22 to 25 for Case A and from 22 to 19 for Case B. For both modifications disk modes did not change and blade-passing frequency remained the same, leaving the number of impeller blades at 17.

In the papers it is shown that triple coincidence occurred as needed for significant response:
1. Tyler / Sofrin spinning mode with 5-diameter as 22-17 = 5
2. 5-Diameter disk mode at 22 X speed
3. Acoustic mode of the gas with 5 diameters = 17X

In summary, coupling of matching modes, acoustic and structural, is necessary at traveling wave resonance at operating speed. A triple coincidence is needed, the same structural and acoustic mode shape, along with the correct number of stationary vanes enabling Tyler / Sofrin spinning acoustic modes, again with the same shape.

There thus is conclusive proof that rotating blade passing frequency does not give concern if it matches frequency of a disk mode. Rather noise excites at traveling wave frequency, disk diameter mode plus or minus number of diameters times speed. Whenever there is an impeller failure especially for a high-pressure "dry" gas compressor acoustic analysis of gas modes at the sides of the cover and hub disk should be considered especially when the two other factors for "triple coincidence" are met. The acoustic analysis described is similar to the disk interaction resonance equations where phase must be matched. The fluctuating pressure mode shape is a function of number of rotating blades and number of stationary vanes. Thus in certain cases there can be a double cause of exciting the same disk mode – acoustic pressure pulsations spinning at blade passing frequency, adding to direct pressure loads due to stationary vanes. The question is answered – high noise at blade passing frequency can rarely cause impellers to fail. Moreover blade-passing frequency is definitely not a concern if it is equal to a disk mode frequency, as long as number of stationary vanes is a number that does not equal rotating blade number, which would be a crucial design error.

CASE STUDIES

". . . unless you do something wrong, it's hard to make changes . . .," Lt. Col. Keith "Sully" Sullivan. For bladed disk resonance, the ramifications of a fracture for a customer can be tremendous. OEMs generally apply "proven" methods to ensure success and minimize failures. However, designs evolve, envelopes are pushed, and when pushed enough, a failure will emerge. A large part of the improvement process is to learn from a limited number of mistakes to arrive at a new "proven" approach. And then the pattern is

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repeated. In every RCFA, there is considerable pressure to arrive at a root cause and we caution against the temptation to grasp hold of some explanation that stretches the imagination a bit too far. The following case studies and accompanying references hopefully will assist the reader.

Axial Compressors

Problems with blade resonance for industrial axial compressors should be less than those encountered by axial turbines. That is, risk can be less if design and operation minimizes rotating stall, and of course a proper surge control system is used to eliminate heavy repetitive surging. Continuous operation at extremely high flow producing stage choking is also to be avoided. There are several prime design issues:

- Ensure that first stage blades avoid very low harmonic excitation—dictated not only by loading (Goodman factor), but by how well the inlet design minimizes circumferential flow distortion;
- Avoid resonance of fundamental modes with upstream vane passing frequency;
- Aerodynamic design must be free of rotating stall at all operating points including those with variable guide vane settings.

Others in the past have described failures with a poor inlet design without proper straightening vanes, followed by adjustable prewhirl vanes. These vanes assist in minimizing non-uniform flow; the prewhirl vanes also provide flow equalization besides giving adjustable flow angles required by the first rotating row. Proper operation requires that variable vanes be closed at startup to eliminate rotating stall. By retaining subsonic designs, conservatism is ensured; transonic designs are much more demanding as relative tip speed increases. Modern designs have more need to have design procedure to avoid blade flutter of first stage blades, high-pressure blades in choke, and acoustic resonance interaction. These factors are reasons why many industrial gas turbines changed to 12% chromium steel with added material damping, as compared to jet engines that must not surge. Inlet air can contain contaminants, thus coatings may be needed for the front stages. Instead of old National Advisory Committee for Aeronautics (NACA) circular-arc blades, using controlled diffusion airfoils (CDA) blades that are actually designed for transonic conditions greatly assists in not exceeding limits. Improvements by using CDA blades are described by Kilgore (1987).

Case A-1, Resonant Operation (Kushner, 2004)

An axial flow air blower had a first row blade resonance with upstream vanes that caused a fatigue crack in the cantilevered blades. The simple solution was to change the number of stationary vanes. There were two lessons learned: (i) A distance of one chord length from stator vanes to rotating blades was not sufficient to kill off the wake stimulus, which was a general rule of thumb in the 1970s, and (ii) using an existing part for an older model must be reviewed for interaction with components in the new design.

Case A-2, Excessive Surge (Kushner, 2004)

A 12-stage axial air compressor in FCC service experienced numerous blade and vane failures that were due to excessive and continuous surging that was also damaging the variable guide vane linkages, and also yielding the casing due to transient thermal stress. The blowoff valve was so undersized that a second blowoff valve had to be added. In addition, the surge control system did not respond adequately to keep the unit out of surge. Strain gages applied in the field verified that acoustic excitation from vortex shedding was not the cause of the failures.

Centrifugal Compressors

Centrifugal stages tend to have lower stresses than their axial counterparts, fewer blades, and better survival to foreign object ingestion. Covered impeller wheels tend to be more robust than open impeller wheels. But failures do still occur. Just like axial compressors, surge and choke operation should be avoided to the fullest extent possible. Industrial centrifugal compressors almost always avoid disk critical speeds in their continuous operating speed ranges, especially for stages with 90-degree inlets and/or volutes.

Case B-1, Open Impeller Disk Critical Speed (Kushner, 2000)

A plant air compressor experienced a failure of an open impeller due to operation at the ND-4 disk critical speed. It requires fairly thin impellers to have disk critical speeds in a typical operating speed ranges, even for open impellers. The impeller was modified by proper thickening and tapering of the disk. A comparison of the initial and final designs (Figure 40) shows how the disk critical speed was shifted upward away from operation. The only stationary vanes were 15 diffuser vanes that did not have critical interaction resonance with any of the primary modes at the operating speed.

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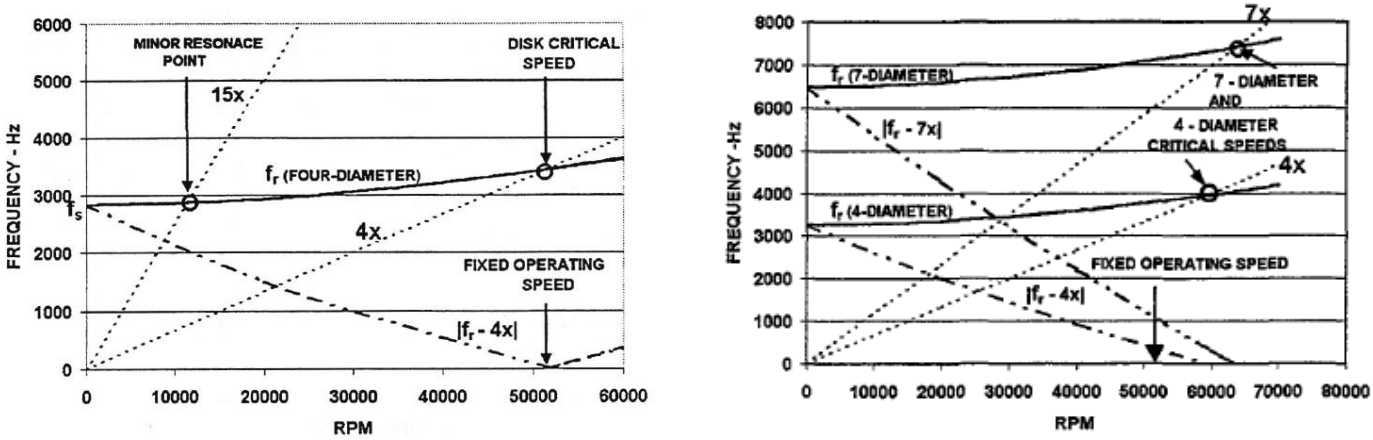

(A) Original (B) Revised

Figure 40. Campbell Diagram for Open Impeller, Removing Disk Critical Speed

Case B-2, Rotating Stall (Kushner, 2004)

A highly loaded air compressor with an open impeller wheel experienced 4x rotating stall excitation when approaching surge. The volute was properly sized based on analysis. The impeller was redesigned using 3-D curved blades and zig-zag pins between the blade leading edge tips similar to those shown in Figure 41.

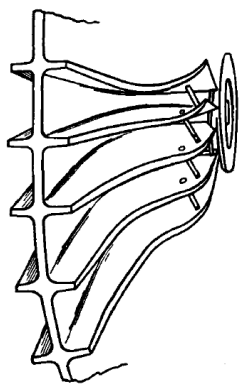

Figure 41. Zig-zag pins applied to an open impeller (Voysey, 1945).

Case B-3 Liquid Ingestion Excites the 2S-B Mode (Kushner, 2000)

The damage of many impellers in four identical refrigeration compressors occurred in the 1970s. Scalloping of the impellers was performed without success. The original thinking at the time was that the cause of the failures was due to excitation of the 3-diameter mode as indicated by the Campbell diagram (Figure 42). However, strain gage instrumentation did not confirm this mode as getting excited in the operating speed range. The strain gage measurements showed that the mode that was actually being excited was the 5-diameter mode. This mode was actually not that excitable except under conditions of liquid ingestion and surging.

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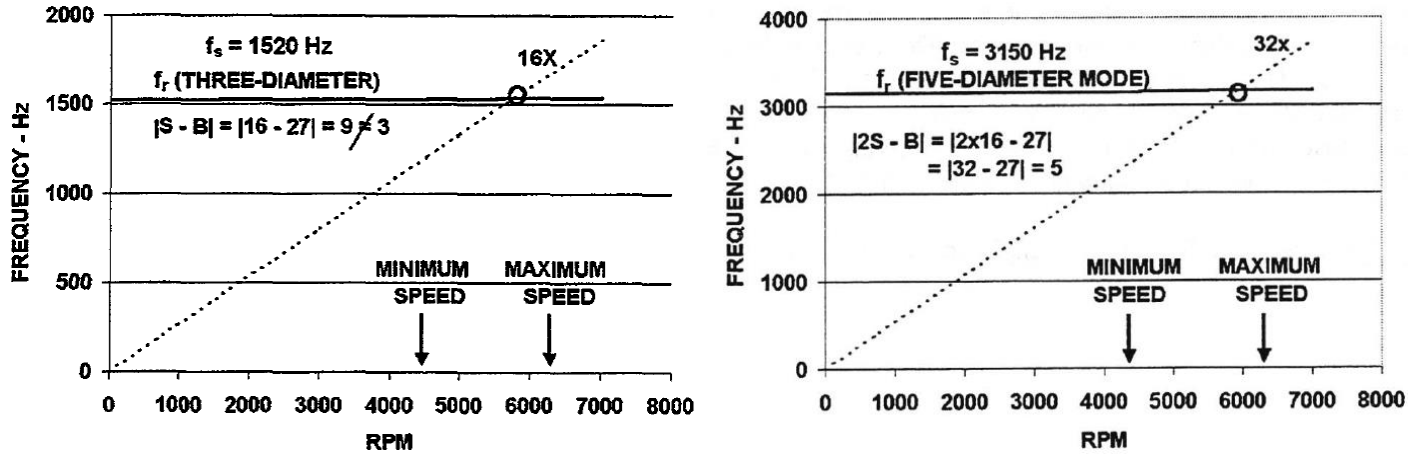

Figure 42. Campbell diagrams of the 3-diameter and 5-diameter modes especially near conditions of surge.

An interference diagram is added in Figure B4 showing that the ND3 mode is not excitable, but the ND5 mode is. Liquid ingestion was the cause and likely the reason for some other gas compressor impeller failures throughout history. For this case strain gage data for field tests were conclusive. With or without liquid ingestion, the three-diameter mode did not respond at its resonance point. The interference diagram (Figure 43) and Equation (8) agree that it cannot be excited.

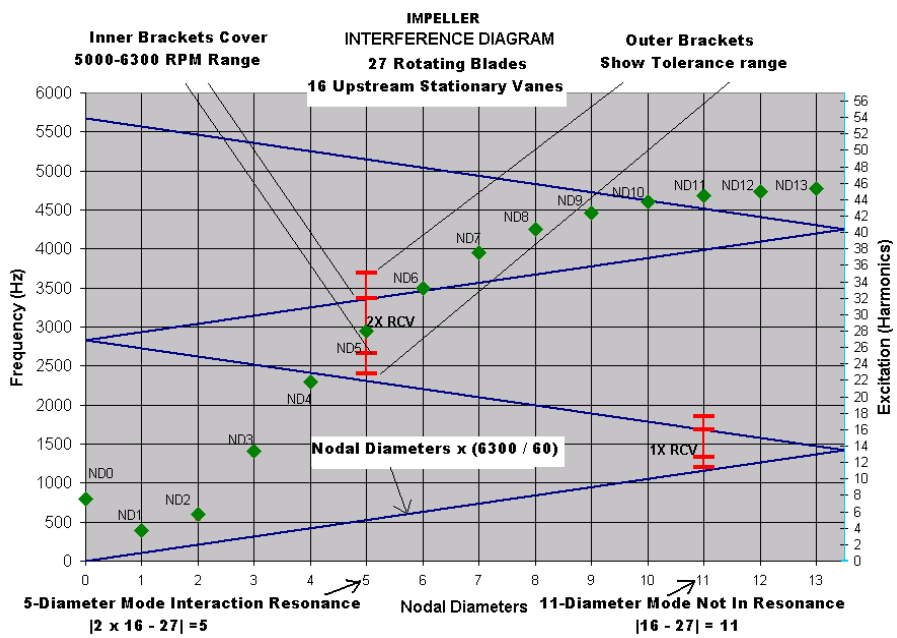

Figure 43. Interference diagram for impeller with 27 blades and 16 upstream vanes.

Case B-4, Impeller Flutter Due to Operation in Choke with Heavy Liquid Ingestion (Kushner, 2000)

A case not explained by interaction resonance caused cracks at the impeller eye and initiation at the toe of the blade welds. The impeller had an unstable response for the fundamental two-diameter mode. In this case operation of the same compressor as described in Case B-3 had an overload operating condition with an erroneous opening of the recycle valve at high flow. In addition there was liquid ingestion. Strain gage measurements indicated impeller flutter whenever heavy liquid ingestion occurred. Centrifugal compressors are easier to design from the stability standpoint as they do not have steep head rise to surge characteristics as do axial compressors affecting aerodynamic damping. One would look at the structure of an impeller compared to a single long axial compressor blade and never surmise that the entire disk would flutter as it did for the 2-diameter disk mode.

Case B-5, Impeller Failure Due to Operation in Choke (Kushner, 2004)

Following a rerate, a large blast furnace air blower suffered several fatigue failures after many years of service. Instead of the reason given by Phillips, et al. (2003), other data showed that the most likely cause was operation in deep choke when the blowoff valve was opened at high flow. As this case does not involve liquids, it is perhaps similar to that described by Borer, et al. (1997), overloading of

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a compressor when a parallel unit tripped offline, driving the running unit into deep choke.

Case B-6, Impeller failure due to unknown cause with increased susceptibility from cutback (Kushner, et al., 2000)
An impeller with cutback blades failed in its first day of operation (Figure 44). A small section of the impeller hub actually tore off, giving high rotor vibration initiating a trip. All potential resonances of primary modes had considerable separation margin; however, FEA and rap testing indicated potential resonance of several plate modes where the blades had been cutback. A review of the process data indicated some readings that did not make sense. The instrumentation was checked, and it was determined that the measured flowrates and the compressor discharge temperature were incorrect. In addition, the process gas had higher density than expected. The plant configuration did not have demisters and liquid ingestion was also suspected. Correction of the flowrates indicated that the actual flowrate was 15% higher than the reading and pushed 5% beyond the end of the curve. However, the end user experience was that operation in or near choke is not normally detrimental and cutbacks are also not detrimental. There were several potential causes that were considered: (i) resonance with inlet vanes of a complex plate mode, (ii) operation near choke with further complications due to potential liquid ingestion and blade cutback. The disk of the spare rotor was machined down and the instrumentation was corrected. The stage did not experience another failure.

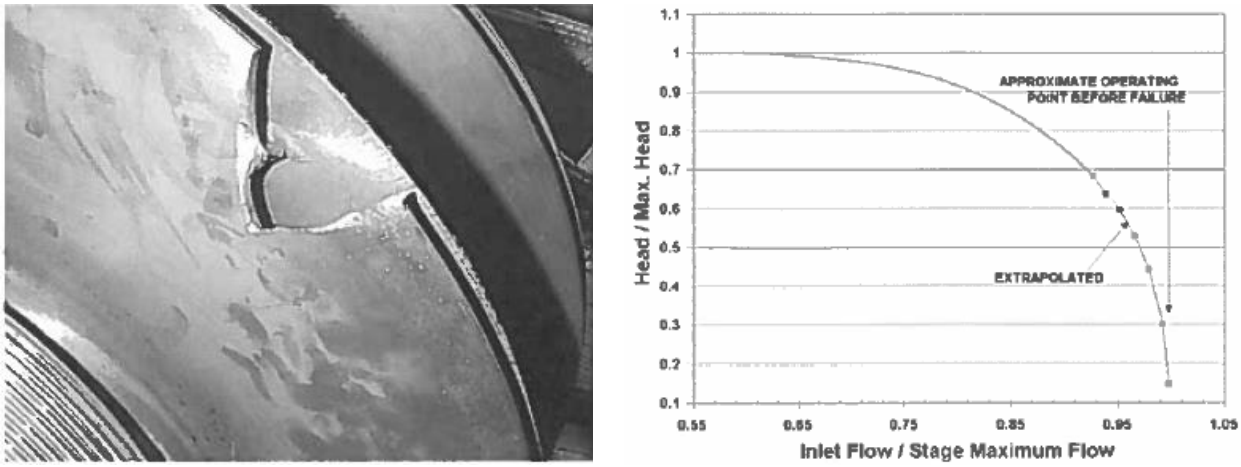

Figure 44. Failed cutback impeller after operating past the end of the curve with likely liquids

Case B-7, Impeller failure due to a cavity gas mode (Kushner, 2004)
Impeller plate modes are sometimes involved in other failures that do not receive intense root cause failure analysis. This case with 19 impeller blades and 16 upstream vanes was reanalyzed to show that impeller failures did not have the triple coincidence of 3-diameter modes for acoustic pulsations, as explained by Petry et al. (2010). Experiencing recurring failures as was this case also is sometimes a reason to have a quick fix. There was no interaction resonance with 16 inlet vanes of the 3-diameter mode; resonance was below minimum operating speed. It is now concluded that rotating blade passing frequency (19 times speed) excited a gas mode on the sides of the impeller, with phase matching of either one of two disk modes: a) a higher frequency 3-diameter / 1-circle mode; or b) a “scallop” plate mode of the impeller. An impeller scallop mode is described in Schiffer and Syed (2006). The recommendation then was to scallop out sections of the impeller hub and cover as shown in Figure B5 (aerodynamic effects are fairly small).

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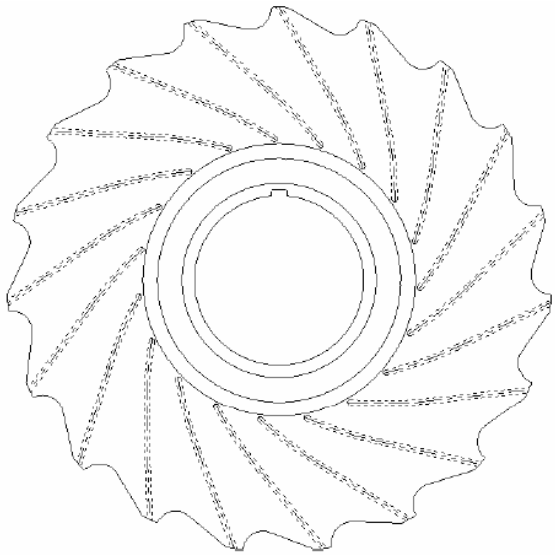

Figure B5. Scalloped hub and cover; 19- bladed impeller

Scalloping increases frequencies of all disk modes, reduces static and dynamic stress, and also modifies acoustic mode frequencies at the sides of the hub and cover. It is concluded that acoustic pulsation phase matching is rare; for this case hydrogen gas had a high sonic speed and thus high gas mode frequencies.

Case B-8 (Kushner, 2004)

High flow compressors employing covered impellers have the most risk of resonance of upstream vane passing frequency with fundamental blade leading edge modes. For this impeller, potential resonance was found for a new unit with the inlet already designed and manufactured. Resonance was below minimum speed and well away from specified operating points so it was considered acceptable. In such cases where resonance is below the minimum speed, it is important to review the startup procedure to ensure that excitable natural frequencies do not coincide with startup hold points.

Case B-9 Demister improvement eliminates impeller failures (Gresh, 2006)

In other impeller fatigue cases besides Cases B-3, B-4, and B-6, some have questioned how liquid ingestion could cause failures. It is well known that liquid slugging can be a major issue, with need for upstream separators / demisters. Also low levels of oil wash sprays are acceptable (at least one case had streams of liquid instead of spray nozzles and failed downstream blades). It does appear to be rare but there is no doubt for the strain gage data proof for Cases B-4 and B-5 above where there were very heavy hydrocarbons liquids in the mixed refrigerant. But there can be effect of two-phase flow for single component gases such as propane, with resolution only after improving separator demisters, e.g. see Gresh (2006). In research for wet gas compressors, Gruner et al. (2008) describing compressor airfoil testing showed that two-phase flow has flow separation and affects sonic velocity. The authors state: “The findings of these experiments and the reviewed literature indicate that a revised compressor design accommodated to wet gas compression is required.” Their findings show liquids can affect excitation especially at poor incidence angles near surge and choke, and thus reduce aerodynamic damping. Two-phase flow reducing sonic speeds could cause earlier choke.

Case B-10 Demister improvement eliminates impeller failures

Another case for a propane compressor shown in Figure 45 where demisters were undersized and modified is described in White et al. (2011). Extensive Fluid Structure Interaction (FSI) analysis described in Lerche et al. (2012) showed that even at excessive flows the predicted response was very low. The covered radial impeller has an integral shroud with different coupling of blades and disks compared to typical axial flow disks. Pettinato, et al. (2013) show comprehensive modal testing using travelling wave excitation with speakers, and mistuning response calculations of a compressor full-inducer covered impeller. Results showed only 0.01% mistuning for a 3-diameter / 0-circle mode of concern with potential interaction of 16 vanes and 13 blades: $|16 - 13| = 3$. Conclusion is there was very low mistuning in a six-diameter pattern. Values for coupled blade leading edge modes gave 0.50% mistuning with calculated possible maximum amplification of 1.3 times a tuned condition. That prediction was based on methods by Feiner and Griffin (2004a; 2004b). Test results verified there were no design issues for the HCF failures of the full inducer impeller design, also described in Kushner et al. (2013) and White et al. (2011). The 3-diameter disk and blade leading edge mode had well over 10% margin from EO 16, upstream vane passing frequency.

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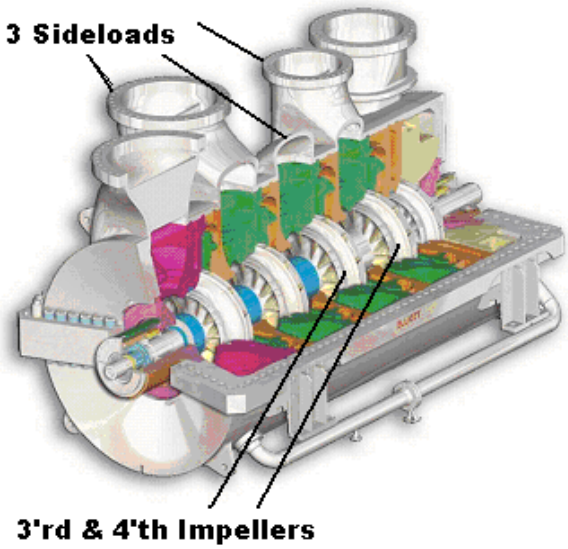

Figure 45. Compressor with Impellers having 13 blades, 16 upstream return channel vanes, and 14 sidestream vanes.

The root cause was concluded to be extreme excitation from liquid ingestion in overload among other potential causes. Operation was corrected with improved demisters added; impeller redesign to improve blade leading edge modes; and sideload redesign to reduce harmonics.

Speed was approaching resonance but with much margin for the mistuned blade modes. A fluid structure interaction (FSI) analysis was conducted. Figure 46 provides the estimated dynamic stress for one of the modes. Response could have been high if it had been at resonance at 4400 rpm, well above maximum design speed, but is very low at actual speed. In the actual case, mistuned individual blade modes would have higher response especially at overload with liquids. FSI results summarized in Figure 47 for a tuned impeller at maximum speed showed dynamic strains in a 3-diameter, not a 1-diameter pattern. Non-uniform inlet flow at the sideload stages added harmonic excitation. The 1-diameter cover/blade torsional modal frequency, near EO 16, was originally concluded to be the responding mode by a consultant. However, the EO-16 would excite the 3-diameter mode, not the 1-diameter mode. There was potential for 1-diameter excitation at EO-14, coinciding with the sideload guide vanes, but this had good separation margin. It should be noted that there are numerous applications with a difference of one between blades and vanes having 1-diameter mode interaction resonance without failure. In Elliott Group's experience, one-diameter modes are coupled to the rotor that provides added damping. There could be blade leading edge modes in a one-diameter pattern that could be concerning.

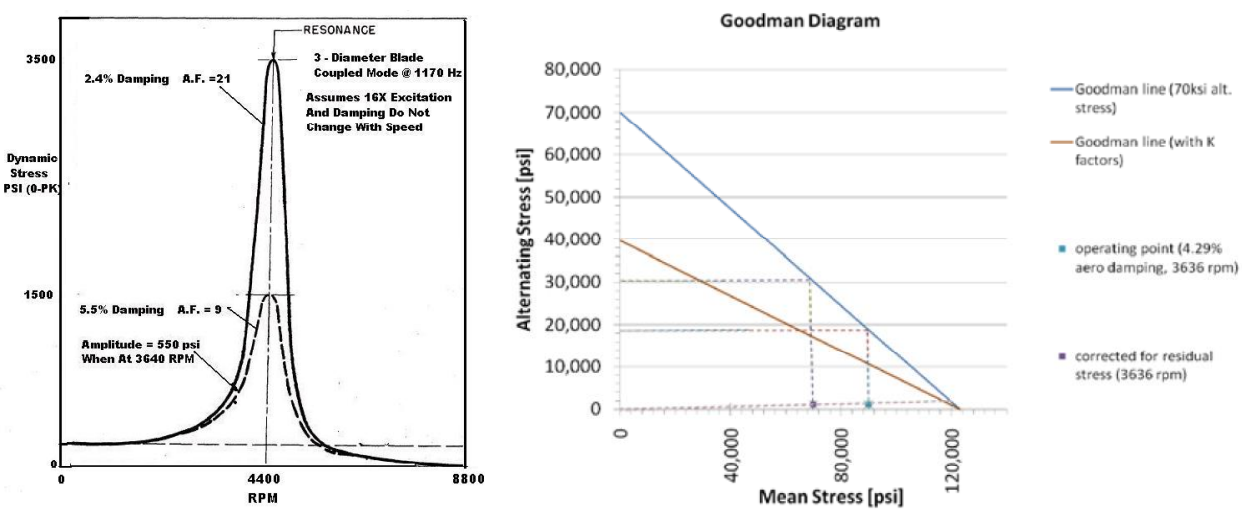

Figure 46. Low response assuming tuned 3-diameter blade coupled mode and Goodman diagram showing very high safety factors.

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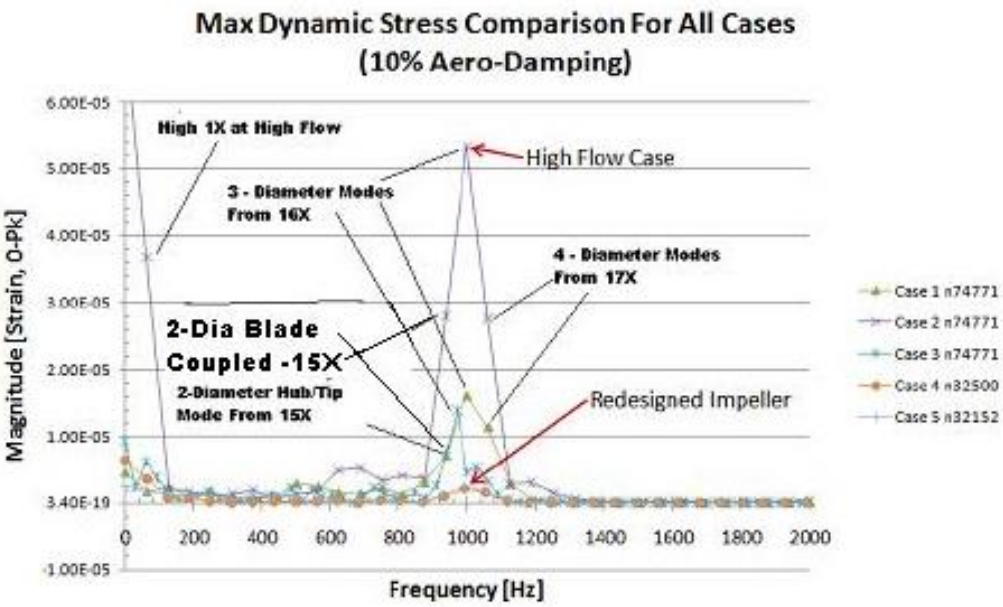

Figure 47. Fluid Structure Interaction Results for Various Cases To Scale From 10% Aerodynamic Damping.

There also was no infringement of design tolerance for resonance or blade leading edge modes, either tuned coupled modes, or for individual mistuned modes. It would require much higher flow and liquid ingestion greatly reducing aerodynamic damping to initiate cracks at the blade leading edges. Along with demister improvements, the third and fourth stage impellers were improved to have even more margin from resonance using same blade and vane numbers. Upstream sideloads were improved to reduce non-uniform flow excitation due to 90-degree casing entries (Figure 48). Blade leading edge modes had higher fundamental leading edge modes with "T-head" design in Figure 49.

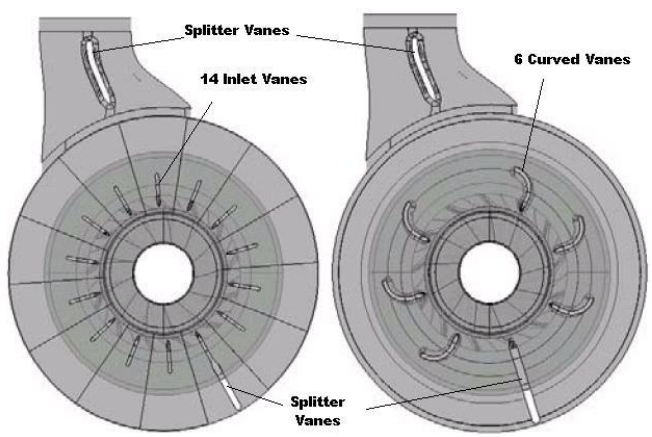

Figure 48. Original (Left) and Revised Sideload Designs.

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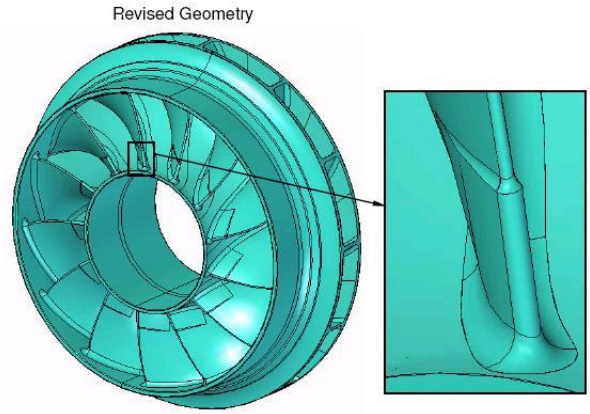

Figure 49. Revised Impeller With T-Head Welds Giving Higher Margin For Blade Mode Resonance

All three upstream sideloads were improved to reduce non-uniform flow excitation due to 90-degree casing entries (Figure 48). Along with sideload demister improvements to reduce misting the third and fourth stage impellers were improved to have even more margin from resonance of both disk and blade modes, using the same impeller blade and return channel vane numbers. Limits in operation were placed to be sure there was not infringement past maximum allowable flow. Rotor inspection during a 2013 planned plant turnaround after two years operation verified the redesign was successful. The customer then rerated the entire plant where the propane compressor impellers were modified for higher flow, retaining the improved welds and shot-peening for added reliability.

Case B-12, Quadruple Coincidence of Acoustic & Flow Excitation (Kushner, 2018)

A standing acoustic wave in the cavity will give alternating forces at the acoustic mode frequency on the sides of a bladed disk. The pulsations act as 360-degree continuous shakers over the entire circumference exciting one of the dual mistuned modes at backward or forward frequencies relative to the rotor. There is a fourth factor; i.e. can be said to be a fourth coincidence for exciting the same disk mode – direct pressure loads due to wakes and potential flow of the same stationary vanes acting on the rotating blades. The question is: are both sources of excitation in-phase? Can there be two modes involved at or near the same speed? Besides reflections for acoustic modes there is interaction of acoustic waves at BPF with the non-uniform flow due to upstream wakes. Also, response to the pressure pulsations spinning at BPF is affected by the swirl within the cavities at the hub and cover side. This may explain why such impeller failures are rare. It would require an FSI – interaction with viscous flow, potential flow effects, damping and acoustics of the entire stage and two nearby modes - to determine phase differences between the two sources.

Triple coincidence includes the same structural and acoustic mode shape, along with the correct number of stationary vanes to enable Tyler - Sofrin spinning acoustic modes at BPF, exciting the cavity acoustic mode. There thus is conclusive proof that rotating BPF is not a concern if it matches frequency of a disk diametral mode as some have claimed. Rather, noise at BPF excites at traveling wave frequency of one of the dual disk diameter modes, plus or minus number of diameters times speed. Depending on spinning mode direction, damping can be higher or lower. Whenever there is an impeller failure, especially for a high-pressure “dry” gas compressor, acoustic analysis of gas modes at the sides of the cover and hub disk should be considered in root cause failure analysis, especially when the two other factors for “triple coincidence” are met. The acoustic analysis described is similar to the disk interaction resonance equations where phase must be matched. The fluctuating pressure mode shape is a function of the number of rotating blades and the number of stationary vanes. The question is answered – high noise at BPF can excite bladed disks, but for complex phase effects rarely causes impellers to fail.

Papers cited on “triple coincidence” do not give details on operation of the compressors that may have been the true cause as other similar compressors with the same triple coincidence have not failed. That agrees with the authors’ experience; in one case there was no doubt that there was extensive operation near and in surge, giving higher noise at BPF and affecting damping. Other similar cases to flow excitation such as given by Kushner et al. (2013) and Ju et al. (2015) could have had too high blade incidence angles at or approaching choke (overload). BPF noise levels increase for flows at the extreme ends of compressor performance maps. That was the case where during first startup of a process compressor using nitrogen for dryout the conditions were such that operation was at the verge of surge for long periods. In Figure 50 is shown that the 5-conditions were met for combined acoustic and flow resonance. The number of vanes was changed to 18 even though there were many such designs with 20 vanes operating at prescribed conditions.

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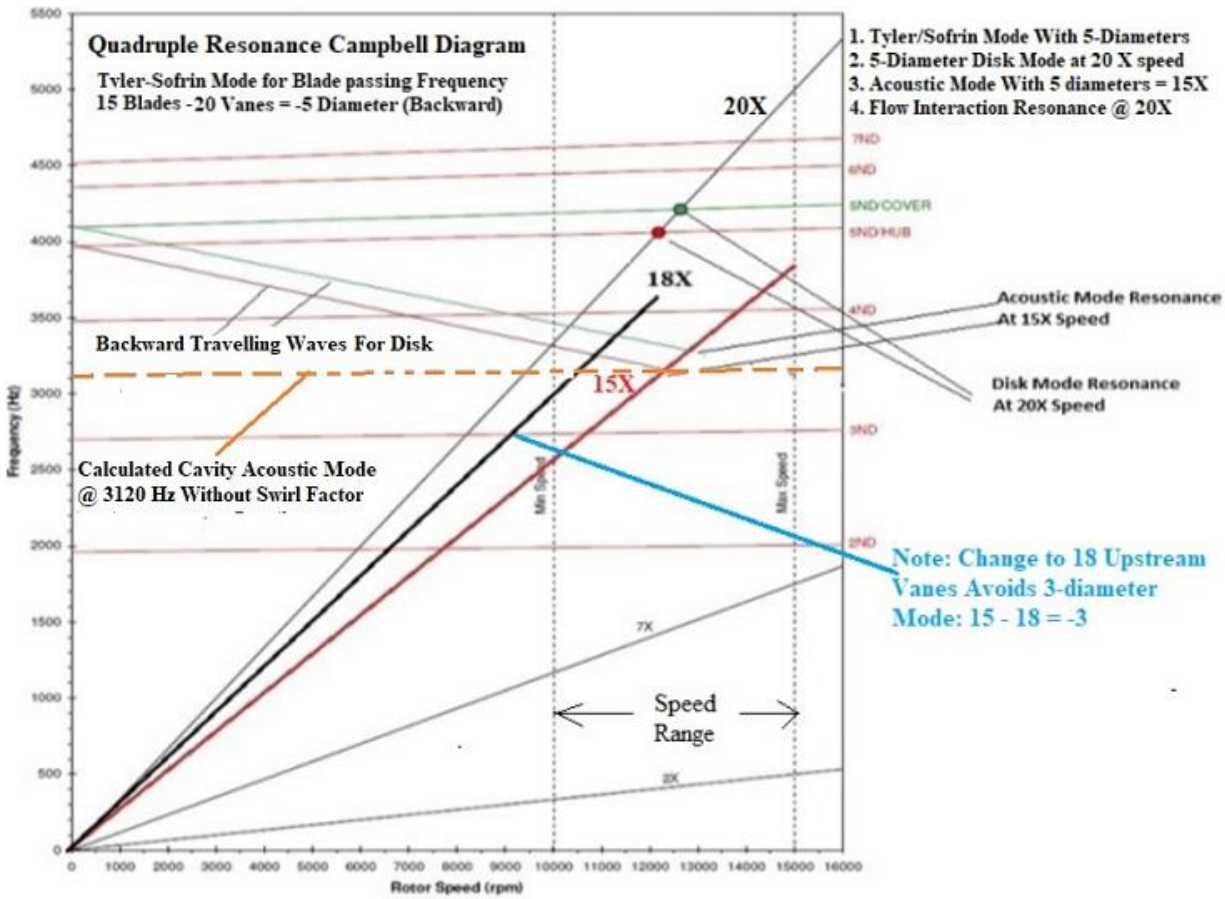

This diagram explains coincidence resonance possible for two different 5-diameter modes, the first mode and a hub/tip mode. The triple coincident acoustic excitation at 15X, blade passing frequency is at sides of the impeller. The flow excitation excites the disk mode at 20 times speed through interaction with rotating blades. Acoustic excitation can be either in or out of phase with disk motion in the axial direction on either side of the impeller. The relative phase angles could also be different for flow Vs acoustic source. In this case change was to 18 vanes that avoids the 3-diameter mode direct flow excitation.

Figure 50. Campbell Diagram with Acoustic Resonance at 15X Near Same Speed as Flow Resonance with 20X

Turbines

Radial-inflow turbines with non-shrouded impellers have similar concerns to compressors, but inlet vane excitation can be worse as to excitation of disk modes due to their location - near the compliant impeller tip. For axial flow steam turbines, as related in a lecture by the Den Hartog (1954), a golden rule for cantilevered long blades is never to have the fundamental mode equal to two per rev, especially for turbine generator sets prone to severe electrical impulses. There have been catastrophic failures due to transient torsional excitation at twice line frequency; in effect the torsional vibration of the entire rotor is exciting the base of the blades. Otherwise the other engineering standards for long blades (and others) are made by a myriad of individual vendors with many designs, shrouding, and friction damping methods (refer for example to Sohre, 1975). Fixed-speed turbines can have accurate tuning to avoid the important resonant modes. Most designs still use 12 percent chromium steels, as there is some benefit of hysteretic damping, although steady-state (mean) stress reduces material damping as shown in Kushner (2004).

Some designs are highly scrutinized so that materials such as titanium and other variations of corrosion-resistant stainless steels can be utilized in a specific design or modification for a problem such as those in the transition zone experiencing corrosion fatigue. The risk can be higher for some variables however so that care must be used, such as avoiding flutter conditions where material damping adds to the positive side of the equation. Recently there have been some examples where blade lives have “only” been around 20 years for some long back-end blades employing Zig-Zag damping pins shown in Figure 3 above. Non-uniformity of upstream nozzles can increase with time in-turn increasing forces at low harmonics of speed. In some cases the most economical solutions would be to add more conservatism by employing Z-lock shrouds or integral shrouds with rolled-in lashing wire.

Case C-1, Corrosion fatigue (Kushner, 2004)

A turbine that had reduced endurance strength in the transition zone suffered a failure with an axial mode having first order resonance with special inlet nozzle vanes used to add rigidity to the diaphragm. Both “stress corrosion” and “corrosion fatigue” can become a contentious issue for steam turbines whenever there is a field failure for a stage in the transition zone. Some of these incidents are likely

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the reason the API 612 committee had an extreme specification for the Fifth Edition with respect to resonance. The higher modes with phase changes along the blade are generally safe even in resonance as response is negligible. Thankfully relief was provided in the Sixth edition of API (2005) by showing acceptable stresses as it was omitted in the Fifth Edition.

It is known that corrosion fatigue is a special case of stress corrosion caused by the combined effects of cyclic stress and corrosion (refer to McCloskey, 2002). Turbine metals are not immune from some reduction of its resistance to cyclic stressing if the metal is in a corrosive environment. By far the highest propensity is where the water droplets first form, at the transition stage. Water, metal, and the presence of a corrosive medium such as chlorides are the three required elements. The first two are a given, the third - the corrosive medium - is almost always the argument, e.g., makeup water treatment of high quality as specified, or another source of contamination. In this case, the pitting was just starting; however, corrosion fatigue is often encountered not as a visible degradation of the metal but as a premature failure of a component under cyclic conditions. Using FEA, the crack initiation site correlated with peak stresses for this out-of-phase mode. As there were operating issues, it was best to provide a more rugged design employing integral shrouds with rolled-in lashing wire. A protective coating similar to that in Bauer et al. (2009) was also considered, but in this case a rerate due to plant changes had already been necessary so that improvements in design and operation fit into the plans.

Case C-2, Rouge failures due to low exhaust pressure and poor damping pin placement (Kushner, 2004)
A variable-speed condensing steam turbine last-stage blade using 12 percent chromium steel, with zig-zag lashing pins experienced two "rouge" failures; only one blade cracked on two rotors with long lives between failures, i.e., years apart. The failures occurred at a small stress riser where the blade surface had limited liquid impingement damage near the stellite overlay. In this case, the loss of aerodynamic damping occurred at too low of exhaust pressure. Damping can also be greatly reduced with abnormally high pressure (stall flutter) while at or near design speed. FEA examination determined that the zig-zag lashing pins were along a nodal line and therefore ineffective for certain modes. The improved design fixed the problem though improved operation may have also contributed.

Case C-3, (Kushner, 2004)
A rerated and repaired steam turbine had the next-to-last stage assembled where all packets could not have the same number of blades. This is not an uncommon situation. The customer's concern was alleviated based on calculated resonant response factors.

Case C-4 (Kushner, 2004)
A radial inflow turbine wheel for a microturbine had two different upstream sources of excitation for a fundamental mode of a cantilevered blade using low damping nickel alloy. Due to the different upstream sources, there was potential for 3x, 6x, 12x, and 15x excitation. Modifications removed concern for the responding blade mode identified resulting in successful application.

Case C-5 (Kushner, 2013)
Two identical turbines with 16-inch long last stage blades had run for over 20 years. One of the units had suffered occasional Zig-Zag lashing wire wear problems and subsequent failures. (See Campbell Diagram in Figure 7). It was unknown whether operation had comprised maximum flow allowable with reduced exhaust pressures and potential water droplet erosion. The design had fundamental tangential mode potential resonance with three times running speed. A new design with continuous z-lock shrouding was utilized and had the lowest mode resonant at a much higher harmonic. The first family of coupled modes had the 4-diameter mode above four times running speed as shown in Figure 51.

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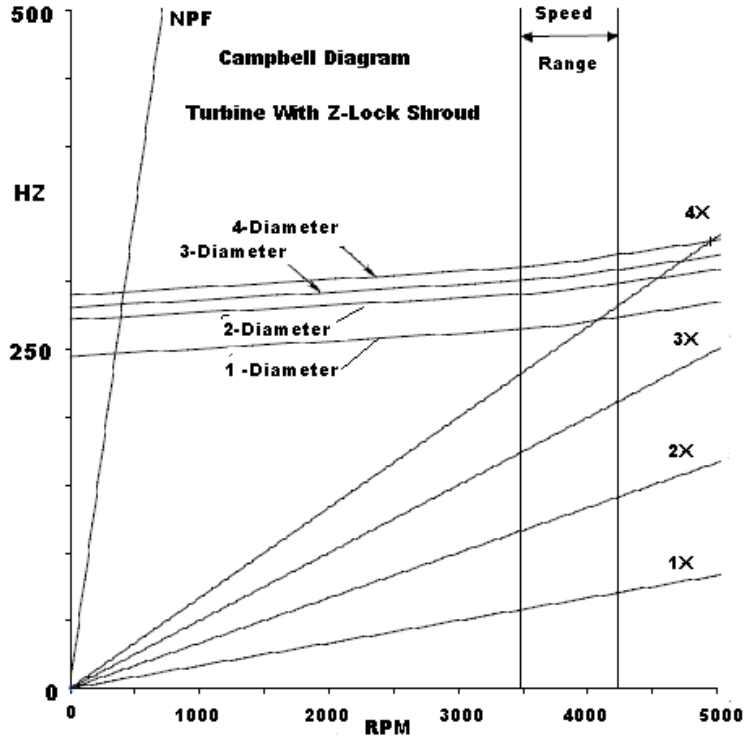

Figure 51. Z-Lock Design With Resonance Above Four Times Speed.

However the new designed blades also failed, likely within just two months based on change in vibration data, where a small piece less than one inch long was ejected from the tip of one blade. The small blade section was found in the exhaust but the decision was made to keep running and the rotor was kept in operation for over five years. The long run was successful as the intact blades still had friction damping contact at their shrouds as they retained their circumferential positions. The failed blade was shorter and likely had its fundamental mode above three times speed. After the long run the spare rotor was installed without any changes. Inspections found that there was foreign object damage that caused a stress riser at the crack initiation point. The blade and two adjacent blades were replaced and the rotor was kept as a spare for the two identical turbines.

Case C-6 (Kushner, 2013)

Rolled in damping wires such as shown in Figure 21 are assembled with clearance between the circumference of the wire and the shroud. At rotational speed, the damping wires bear against the rolled lips of the shroud; and friction damping is provided by relative movement between the wires and the shroud. This case showed that poor steam purity cause severe fouling of the damping wires and shroud which not only disabled the friction damping mechanism of the blade assembly, but also significantly changed the natural frequencies of the blade assembly vibration modes. The blades failed after approximately 5 years in service. The failed blades were located at the transition stage for this condensing steam turbine. A metallurgical examination showed that the failure mode was fatigue with cracks initiating at small surface pits in the blade roots. EDS showed chloride and some sodium corrosive elements in the surface pits; however, corrosion was only a contributing factor in the blade root failures. Optical examination of the damping wire assembly showed that the wires were fouled to the extent of being fixed to the shroud. Finite element modal analysis was used to evaluate a clean blade assembly without damping wires coupled to the shroud and the fouled blade assembly with damping wires coupled to the shroud. FEA results for the clean blade assembly showed potential interference only for a nonresponsive higher order blade bending mode at nozzle passing frequency. Properly functioning damping wires would provide friction damping for this higher order blade bending mode if this mode could be excited at all. FEA results for the fouled blade assembly showed a significant increase in blade assembly natural frequencies with increasing nodal diameter compared to the clean blade assembly. FEA results for the fouled blade assembly showed potential interference of the 1st axial blade mode/60 nodal diameter pattern at nozzle passing frequency (Figure 52). There is always some uncertainty when specifying fixed displacement constraints and coupled degrees of freedom in a finite element modal analysis. For variable speed machines with a wide operating speed range, extreme accuracy may not be necessary to show that a system is probably in an unfavorable condition.

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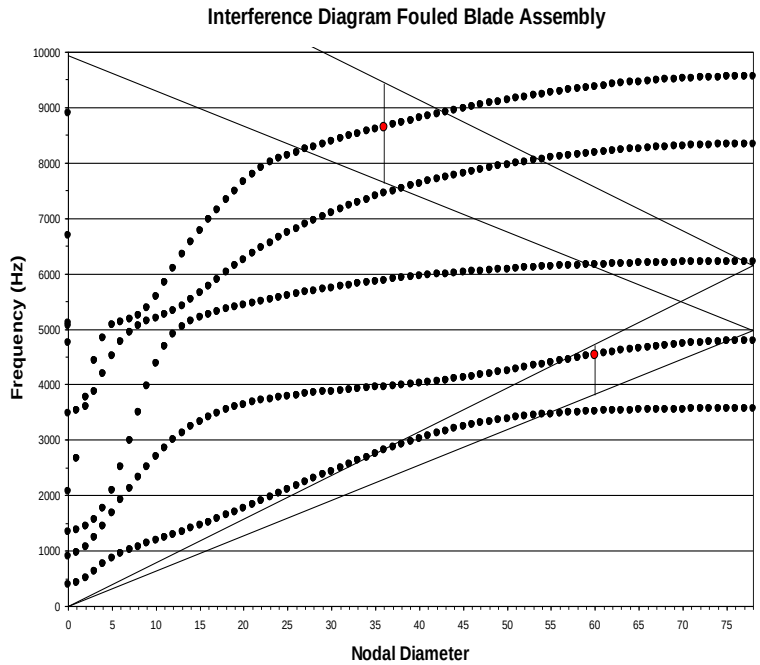

Figure 52. Resonance of first axial mode with 60 nozzles

The blade assembly was modified to be more tolerant to fouling. The blades were assembled in shrouded packets with 10 packets of 13 blades per packet and two packets of 12 blades per packet. A difference in airfoil count resulted from using a locking piece instead of locking blade. In order to reuse the integral disk, oversized locking pin holes were required which could not support a locking blade at rotational speed. Finite element modal analysis results for the assembled blade packets showed no significant interference between blade packet modes and nozzle passing frequencies. With 60 nozzles per circle and 156 blade spaces per circle; resonant response factor for 13 blades per packet is zero, for 12 blades per packet is 0.083 (APPENDIX A). Thus, the RRF is optimized for all blades with in-phase packet modes. Higher modes of the packets also have response factors below 0.20.

CONCLUSIONS

- The methods described using parametric Equations (8) and (9) should be utilized whenever possible, not only for failure analysis but also in the design phase such as in selecting number of vanes for vaned diffusers and blades for all new bladed-disk designs.
- Equal numbers of vanes and blades should always be avoided; often while also reviewing aerodynamic effects, a more optimum design can utilize a difference of one between stationary vane and rotating blade numbers. Even and odd combinations of stationary and rotating blades are usually a better choice.
- Modal tests show that when the blade number divided by the diameter number is an integer, the mode is very clean and responsive. Therefore, a prime number of blades appears to be a better choice to limit the response of such modes.
- Although covered centrifugal compressor impellers are definitely much more rugged than open impellers, first order resonance of the blade leading edge mode can require as much scrutiny as the first few modes for freestanding cantilevered blades in other machinery.
- There is always a need to scrutinize operating and environmental parameters, not only in design specifications, but also in problem resolution.
- Extensive finite element analysis can be used to correlate vibratory modes and location of peak stresses with fracture analysis documenting crack initiation sites for a failed specimen. FSI combining CFD and FEA, may sometimes be required after comparing “is and is nots” in an RCFA.
- Operation of machinery outside of their respective maps is the cause of many failures, both compressors and turbines. These conditions also could present added interaction resonance concerns, especially for high gas density and loads in radial impellers. Running compressors in deep stonewall should be avoided; the manufacturer should review any operation past the end of the allowable curve. If inadvertent high flow condition occurs, limits should be used for transients with limited number of cycles, just as is recommended for surge.

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- Liquid ingestion must also be avoided since it intensifies engine order resonance, and has caused at least one verified case of loss of overall damping. Online liquid monitoring and prevention procedures should be developed for both compressors and turbines.
- Case studies such as those given will hopefully lead to more prevention of inherent design limitations, as well as ensuring proper operation of all fluid-handling turbomachinery for the oil, gas, petrochemical, and related industries.
- Steam turbine transition stages require more attention in both design and material selection for minimizing corrosion. Protective coatings can be utilized to enhance corrosion resistance.
- Because mistuning of blades is unavoidable, it is recommended to use care in replacing phase analyses such as that in APPENDIX A with the graphing method suggested by others in conjunction with nodal diameters, especially for modes of short blades.
- Blades that are locked together by continuous shrouding or by z-shrouds behave as a disk. Interference diagrams using nodal diameters should be employed for these cases where the blades are all coupled together.
- Additional research and implementation of new techniques replacing glass bead shot peening such as laser shock-peening, and low plasticity burnishing as well as new coatings will aid in precluding fatigue failures.
- Increased research and development and advanced Fluid Structure Interaction (FSI) analyses and acoustic / structural checks will get turbomachinery manufacturers closer to optimum designs for high cycle fatigue (HCF) protection. Included in some rare cases will be need for 2-way FSI to obtain more accurate aerodynamic damping predictions.

APPENDIX A

Blade Packet Optimization for the Fundamental Mode

The equation to calculate response factor, R, for a packet of blades where all blades are in-phase with each other and have the same amplitude is:

$$R = \frac{\sqrt{X^2+Y^2}}{p} \qquad \text{(A-1)}$$

where:

R = *Resonant response factor*
X $= \sum_{i=1}^{p} cos[(i-1)\cdot\alpha]$
Y $= \sum_{i=1}^{p} sin[(i-1)\cdot\alpha]$
α = *Relative phase angle, degrees [360 · (s/b)]*
s = *Number of stationary vanes or nozzles*
b = *Number of rotating blades per row*
p = *Number of blades per packet*

These equations can be consolidated for a single equation to more readily find the optimum number of blades per packet, p:

$$J = k \cdot |s-b|/b \qquad \text{(A-2)}$$

Set k = 1, k = 2, k =3, etc., and calculate values of J. Then optimum value of blades per packet, p, is the value used for k whenever J is equal to, or is closest to, a whole integer. Multiples of the optimum value of p can also be used, as long as the shroud can be assembled with higher number of blades per packet. An example is given in Table A-1.

Table A1. Values of "J" for Selected Values of Integer "k" for S = 80, B = 64.

Integer "k"	Value of "J"
1	0.25
2	0.50
3	0.75
4	**1.00**
5	1.25
6	1.50
7	1.75
8	**2.00**

In this case, using k = 4 or 8 results in an integer (not always the case—the closer to an integer the better). The resonant response factor can be verified using Equation (B-1), the response factor for p = 4 or p = 8 would be zero, i.e., perfect phase cancellation. The row of 64 blades could thus be assembled with either 16 packets of four blades, or eight packets of eight blades. Assume that this was a fairly long blade at the transition zone of a steam turbine, and there was no chance of resonance with an in-phase mode. If an out-of-phase

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mode was a cause of an unusual field problem, such as with excessive corrosion, then it could be much better to choose two packets with five blades and nine packets having six blades.

APPENDIX B

Harmonic Excitation - Fourier Analysis

For a pulse train made up of a number of rectangular pulses, m, over 360 degrees (one revolution), to calculate S_h, peak-to-peak exciting force at harmonic, *h*,

$$S_h = \sqrt{C^2 + D^2} \qquad \text{(B-1)}$$

where:

S_h = Peak-to-peak exciting force at harmonic h

$C = \sum_{i=1}^{m} A_i \cdot cos(h \cdot \theta_i)$

$D = \sum_{i=1}^{m} A_i \cdot sin(h \cdot \theta_i)$

where:

h = *Harmonic of speed*

A_m = *Peak-to-peak force of each pulse*

β_m = *Angle from zero degrees to center of each pulse*

and:

$$A_m = a_m \cdot \frac{2}{h \cdot \pi} \cdot \sin\left(h \cdot \frac{\beta_m}{2}\right) \qquad \text{(B-2)}$$

where:

a_m = *Fraction of maximum force of any pulse*

β_m = *Rectangular width of pulse, degrees*

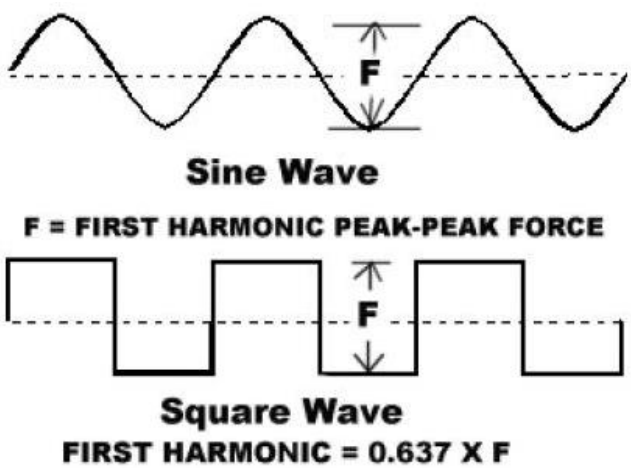

Figure B1. First Harmonic Excitation Factor: Square Wave Compared To Sine Wave

For a sine wave shown in Figure B-1, at resonance, the exciting force is in-phase with the response over the entire vibration cycle. As the force is out-of-phase over part of the cycle, a square wave for example would only have 0.637 times the exciting force of a sine wave at the first harmonic. For the square wave, a rectangular pulse with β = 180 degrees, Equation (B-2) simplifies to:

$$S_h = \left|\frac{2}{h \cdot \pi} \cdot \sin(h \cdot 90°)\right| \qquad \text{(B-3)}$$

First harmonic, $S_1 = |2 / \pi \cdot \sin(90)| = 0.637$

Second, $S_2 = |2 / 2\pi \cdot \sin(180)| = 0.0$

Third, $S_3 = |2 / 3\pi \cdot \sin(270)| = 0.212$

Fourth, etc.

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